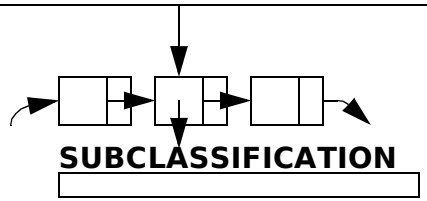
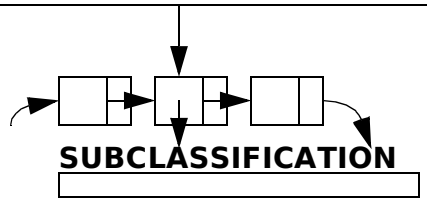
ELEMENT

```
/* Usage */
ELEMENT *element_fetch(
	char *element_name );

/* Process */
char *element_primary_where(
	char *element_name );

char *element_system_string(
	char *ELEMENT_SELECT,
	char *ELEMENT_TABLE,
	char *element_primary_where()
		/* where */ );

ELEMENT *element_parse(
	char *input );

ELEMENT *element_new(
	char *element_name );

ELEMENT *element_calloc( void );

/* Input */
char *element_name;
boolean accumulate_debit;

/* Usage */
LIST *element_subclassification_list(
	char *element_name,
	char *begin_transaction_date_time,
	char *end_transaction_date_time );

/* Process */
LIST *subclassification_list = list_new();

LIST *subclassification_element_subclassification_name_list(
	element_primary_where( element_name ),
	SUBCLASSIFICATION_TABLE,
	“display_order” /* order_column */ );

for subclassification_name in
	subclassification_element_subclassification_name_list()
{
	list_set(
		subclassification_list,
		subclassification_element_fetch(
			subclassification_name,
			begin_transaction_date_time,
			end_transaction_date_time ) );
}

/* Usage */
LIST *element_account_list(
	LIST *subclassification_list );

/* Process */
LIST *account_list = list_new();

for SUBCLASSIFICATION *subclassification in subclassification_list
{
	list_set_list(
		account_list,
		subclassification->account_list );
}

LIST *account_balance_sort_list( account_list );
```



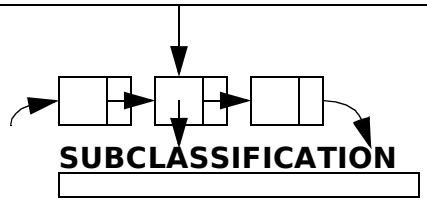
ELEMENT_FUND

```
/* Usage */
ELEMENT_FUND *element_fund_fetch(
	char *element_name,
	char *fund_name,
	char *begin_transaction_date_time,
	char *end_transaction_date_time,
	boolean fetch_subclassification_list,
	boolean fetch_account_list,
	boolean fetch_journal_list );

/* Process */
char *element_primary_where(
	element_name );

char *element_system_string(
	ELEMENT_SELECT,
	ELEMENT_TABLE,
	element_primary_where()
		/* where */ );

ELEMENT_FUND *element_fund_parse(
	char *input,
	char *fund_name,
	char *begin_transaction_date_time,
	char *end_transaction_date_time,
	boolean fetch_subclassification_list,
	boolean fetch_account_list,
	boolean fetch_journal_list );

ELEMENT_FUND *element_fund_new(
	char *element_name,
	char *fund_name );

ELEMENT_FUND *element_fund_calloc( void );

/* Input */
char *element_name;
char *fund_name;
boolean accumulate_debit;
```



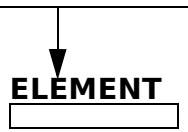
ELEMENT_PROGRAM

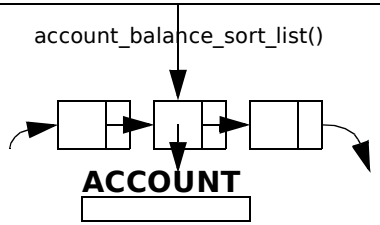
```
/* Usage */
ELEMENT_PROGRAM *element_program_fetch(
	char *element_name,
	char *program_name,
	char *begin_transaction_date_time,
	char *end_transaction_date_time,
	boolean fetch_subclassification_list,
	boolean fetch_account_list,
	boolean fetch_journal_list );

/* Process */
char *element_primary_where(
	element_name );

char *element_system_string(
	ELEMENT_SELECT,
	ELEMENT_TABLE,
	element_primary_where()
		/* where */ );

ELEMENT_PROGRAM *element_program_parse(
	char *input,
	char *program_name,
	char *begin_transaction_date_time,
	char *end_transaction_date_time,
	boolean fetch_subclassification_list,
	boolean fetch_account_list,
	boolean fetch_journal_list );

ELEMENT_PROGRAM *element_program_new(
	char *element_name,
	char *program_name );

ELEMENT_PROGRAM *element_program_calloc( void );

/* Input */
char *element_name;
char *program_name;
boolean accumulate_debit;
```



SUBCLASSIFICATION

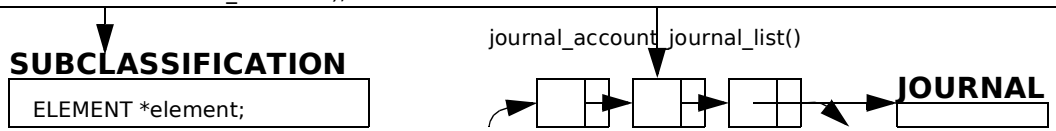
```
/* Usage */
SUBCLASSIFICATION *subclassification_fetch(
   char *subclassification_name,
   boolean fetch_element );

/* Process */
char *subclassification_primary_where(
   char *subclassification_name );

char *subclassification_system_string(
   char *SUBCLASSIFICATION_SELECT,
   char *SUBCLASSIFICATION_TABLE,
   char *subclassification_primary_where()
         /* where */ );

SUBCLASSIFICATION *subclassification_parse(
   char *input,
   boolean fetch_element );

SUBCLASSIFICATION *subclassification_new(
   char *subclassification_name );

SUBCLASSIFICATION *subclassification_calloc(
   void );

/* Input */
char *subclassification_name;
char *element_name;
int display_order;

if ( fetch_element )
{
   ELEMENT *element =
         element_fetch(
               subclassification->element_name );
}

/* Usage */
SUBCLASSIFICATION *subclassification_element_fetch(
   char *subclassification_name,
   char *begin_transaction_date_time,
   char *end_transaction_date_time );

/* Process */
SUBCLASSIFICATION *subclassification =
   subclassification_new(
         subclassification_name );

LIST *account_subclassification_account_name_list(
   ACCOUNT_TABLE,
   subclassification_primary_where(
         subclassification_name ) );

if ( !list_rewind(
   account_subclassification_account_name_list() )
{
   return (SUBCLASSIFICATION *)0;
}

LIST *account_list = list_new();

for account_name in
   account_subclassification_account_name_list()
{
   list_set(
         account_list,
         account_subclassification_fetch(
               account_name,
               begin_transaction_date_time,
               end_transaction_date_time ) );
}

LIST *account_balance_sort_list(
   account_list );

/* Public */
LIST *
   subclassification_element_subclassification_name_list(
         char *element_primary_where(),
         char *SUBCLASSIFICATION_TABLE,
         char *order_column );
```





ACCOUNT (1)

```
/* Usage */
ACCOUNT *account_fetch(
      char *account_name,
      boolean fetch_subclassification,
      boolean fetch_element );

/* Process */
char *account_primary_where(
      char *account_name );

char *account_system_string(
      char *ACCOUNT_SELECT,
      char *ACCOUNT_TABLE,
      char *account_primary_where()
            /* where */ );

ACCOUNT *account_parse(
      char *input,
      boolean fetch_subclassification,
      boolean fetch_entity );

ACCOUNT *account_new( char *account_name );

ACCOUNT *account_calloc( void );

/* Input */
char *account_name;
char *subclassification_name;
char *hard_coded_account_key;
int chart_account_number;
int annual_budget;

if ( fetch_subclassification )
{
      SUBCLASSIFICATION *subclassification =
            subclassification_fetch(
                  subclassification_name,
                  fetch_entity );
}

/* Attributes */
double balance;
double payment_amount;
LIST *journal_account_journal_list;
int percent_of_asset;
int percent_of_revenue;
int delta_prior;
LIST *following_balance_zero_journal_list;
LIST *liability_journal_list;
double liability_due;
char *action_string;

/* Usage */
ACCOUNT *account_key_fetch(
      char *hard_coded_account_key,
      boolean fetch_subclassification,
      boolean fetch_entity );

/* Process */
char *account_hard_coded_key_where(
      char *hard_coded_account_key );

ACCOUNT *account_parse(
      string_pipe(
            account_system_string(
                  ACCOUNT_SELECT,
                  ACCOUNT_TABLE,
                  account_hard_coded_key_where(
                        hard_coded_account_key ) ) ),
      fetch_subclassification,
      fetch_entity );

/* Usage */
LIST *account_system_list(
      char *account_system_string(),
      boolean fetch_subclassification,
      boolean fetch_element );

/* Usage */
LIST *account_list(
      char *where,
      boolean fetch_subclassification,
      boolean fetch_element );

/* Process */
char *account_system_string(
      ACCOUNT_SELECT,
      ACCOUNT_TABLE,
      where );

LIST *account_system_list(
      account_system_string(),
      fetch_subclassification,
      fetch_element );

/* Usage */
ACCOUNT *account_subclassification_fetch(
      char *account_name,
      char *begin_transaction_date_time,
      char *end_transaction_date_time );

/* Process */
ACCOUNT *account =
      account_new(
            account_name );

LIST *journal_account_journal_list(
      account_name,
      begin_transaction_date_time,
      end_transaction_date_time );

/* Usage */
boolean account_accumulate_debit(
      char *account_name,
      boolean expect_lots );

/* Process */
if ( expect_lots )
{
      static LIST *list = account_list( “1 = 1”, 0, 0 );

      ACCOUNT *account =
            account_seek(
                  account_name,
                  list );
      return
      subclassification->element->accumulate_debit;
}
else
{
      ACCOUNT *account =
            account_fetch(
                  account_name,
                  1 /* fetch_subclassification */,
                  1 /* fetch_element */ );

      return
      subclassification->element->accumulate_debit;
}
```



ACCOUNT (2)

```
/* Usage */
LIST *account_balance_sort_list(
	LIST *account_list );

/* Process */
int account_balance_match_function(
	ACCOUNT *account_from_list,
	ACCOUNT *account_compare );

JOURNAL *from_list_latest_journal =
	list_last(
		account_from_list->
			journal_account_journal_list() );

JOURNAL *compare_latest_journal =
	list_last(
		account_compare->
			journal_account_journal_list() );

if ( !from_list_latest_journal
||   !from_list_latest_journal->balance )
{
	return 1;
}

if ( !compare_latest_journal
||   !compare_latest_journal->balance )
{
	return 1;
}

if (   from_list_latest_journal->balance <=
	compare_latest_journal->balance )
{
	return 1;
}
else
{
	return -1;
}

/* Usage */
LIST *account_list_delta_prior_set(
	LIST *prior_account_list,
	LIST *account_list );

/* Process */
void account_delta_prior_set(
	LIST *prior_account_list,
	ACCOUNT *account );

ACCOUNT *prior_account =
	account_seek(
		account->account_name,
		prior_account_list );

JOURNAL *prior_latest_journal =
	journal_list_latest(
		prior_account->
			journal_account_journal_list );

JOURNAL *latest_journal =
	journal_list_latest(
		account->
			journal_account_journal_list );

int account_delta_prior(
	double prior_latest_journal->balance,
	double latest_journal->balance );

/* Usage */
LIST *account_list_percent_of_asset_set(
	LIST *account_list,
	double asset_total );

/* Process */
for ACCOUNT *account in account_list
{
	account->percent_of_asset =
		float_percent_of_total(
			account_balance(
				account->
					journal_account_journal_list ),
			asset_total );
}

/* Usage */
LIST *account_list_percent_of_revenue_set(
	LIST *account_list,
	double revenue_total );

/* Process */
for ACCOUNT *account in account_list
{
	account->percent_of_revenue =
		float_percent_of_total(
			account_balance(
				account->
					journal_account_journal_list ),
			revenue_total );
}
```

ACCOUNT (3)

```
/* Usage */
double account_balance(
	LIST *account->journal_account_journal_list );

/* Process */
JOURNAL *latest_journal =
	journal_list_latest(
		journal_account_journal_list );

return latest_journal->balance;

/* Usage */
double account_debit_total(
	LIST *account_list );

/* Process */
for ACCOUNT *account in account_list
{
	JOURNAL *latest_journal =
		journal_list_latest(
			account->journal_account_journal_list );

	if ( !account_accumulate_debit(
		account->account_name,
		1 /* expect_lots */ )
	&& latest_journal->balance < 0.0 )
	{
		debit_total += -latest_journal->balance;
	}
	else
	if ( account_accumulate_debit(
		account->account_name,
		1 /* expect_lots */ )
	&& latest_journal->balance > 0.0 )
	{
		debit_total += latest_journal->balance;
	}
}

/* Usage */
double account_credit_total(
	LIST *account_list );

/* Process */
for ACCOUNT *account in account_list
{
	JOURNAL *latest_journal =
		journal_list_latest(
			account->journal_account_journal_list );

	if ( account_accumulate_debit(
		account->account_name,
		1 /* expect_lots */ )
	&& latest_journal->balance < 0.0 )
	{
		credit_total += -latest_journal->balance;
	}
	else
	if ( !account_accumulate_debit(
		account->account_name,
		1 /* expect_lots */ )
	&& latest_journal->balance > 0.0 )
	{
		credit_total += latest_journal->balance;
	}
}

/* Usage */
double account_balance_total(
	LIST *account_list );

/* Process */
for ACCOUNT *account in account_list
{
	JOURNAL *latest_journal =
		journal_list_latest(
			account->journal_account_journal_list );

	balance_total += latest_journal->balance;
}

/* Usage */
char *account_hard_coded_account_name(
	char *account_key,
	boolean warning_only,
	const char *calling_function_name );

/* Process */
ACCOUNT *account_key_seek(
	char *account_key,
	static LIST *account_list() );

/* Usage */
LIST *account_subclassification_account_name_list(
   char *ACCOUNT_TABLE,
   char *subclassification_primary_where() );

/* Process */
char *account_subclassification_account_system_string(
	char *ACCOUNT_TABLE,
	char *subclassification_primary_where() );

LIST *pipe2list(
	account_subclassification_account_system_string()
	);
```

ACCOUNT (4)

```
/* Usage */
void account_list_action_string_set(
        LIST *account_list,
        char *application_name,
        char *session_key,
        char *login_name,
        char *role_name,
        char *beginning_date,
        char *as_of_date );

/* Process */
char *account_action_string(
        char *application_name,
        char *session_key,
        char *login_name,
        char *role_name,
        char *begin_date_string,
        char *as_of_date,
        char *account_name );

/* Public */
char *account_name_format(
        char *account_name );

char *account_name_display(
        char *account_name );

ACCOUNT *account_seek(
        char *account_name,
        LIST *account_list() );

char *account_depreciation_expense(
        char *ACCOUNT_DEPRECIATION_KEY );

char *account_accumulated_depreciation(
        char *ACCOUNT_ACCUMULATED_KEY );

char *account_sales_tax_payable(
        char *ACCOUNT_SALES_TAX_PAYABLE_KEY );

char *account_revenue(
        char *ACCOUNT_REVENUE_KEY );

char *account_receivable(
        char *ACCOUNT_RECEIVABLE_KEY );

char *account_payable(
        char *ACCOUNT_PAYABLE_KEY );

char *account_uncleared_checks(
        char *ACCOUNT_UNCLEARED_CHECKS_KEY );

char *account_shipping_revenue(
        char *ACCOUNT_SHIPPING_REVENUE_KEY );

char *account_cash(
        char *ACCOUNT_CASH_KEY );

char *account_gain(
        char *ACCOUNT_GAIN_KEY );

char *account_loss(
        char *ACCOUNT_LOSS_KEY );

char *account_fees_expense(
        char *ACCOUNT_FEES_EXPENSE_KEY );

char *account_sales_tax_expense(
        char *ACCOUNT_SALES_TAX_EXPENSE_KEY );

char *account_closing_entry(
        char *ACCOUNT_CLOSING_KEY );

char *account_drawing(
        char *ACCOUNT_DRAWING_KEY );

LIST *account_cash_name_list(
        char *ACCOUNT_TABLE,
        char *SUBCLASSIFICATION_CASH );

LIST *account_current_liability_name_list(
        char *ACCOUNT_TABLE,
        char *SUBCLASSIFICATION_CURRENT_LIABILITY,
        char *ACCOUNT_UNCLEARED_CHECKS );

LIST *account_receivable_name_list(
        char *ACCOUNT_TABLE,
        char *SUBCLASSIFICATION_RECEIVABLE );

ACCOUNT *account_getset(
        LIST *account_list,
        char *account_name );
```

JOURNAL (1)

```
/* Usage */
LIST *journal_year_list(
	int tax_year,
	char *tax_form_line_account->account_name );

/* Process */
char *journal_year_where(
	int tax_year,
	char *account_name,
	char *PREDICTIVE_PRECLOSE_TIME );

LIST *journal_system_list(
	journal_system_string(
		JOURNAL_SELECT,
		JOURNAL_TABLE,
		journal_year_where() ),
	0 /* not fetch_check_number */,
	1 /* fetch_memo */ );

/* Usage */
LIST *journal_minimum_list(
	char *minimum_transaction_date_time,
	char *account_name );

/* Process */
char *journal_minimum_where(
	char *minimum_transaction_date_time,
	char *account_name );

LIST *journal_system_list(
	journal_system_string(
		JOURNAL_SELECT,
		JOURNAL_TABLE,
		journal_minimum_where() ),
	0 /* not fetch_check_number */,
	0 /* not fetch_memo */ );

/* Usage */
LIST *journal_entity_list(
	char *full_name,
	char *street_address,
	char *account_name );

/* Process */
char *journal_entity_where(
	char *full_name,
	char *street_address,
	char *account_name );

LIST *journal_system_list(
	journal_system_string(
		JOURNAL_SELECT,
		JOURNAL_TABLE,
		journal_entity_where() ),
	0 /* not fetch_check_number */,
	0 /* not fetch_memo */ );

/* Usage */
JOURNAL *journal_account_fetch(
	char *transaction_date_time,
	char *account_name );

/* Process */
char *journal_transaction_account_where(
	char *transaction_date_time,
	char *account_name );

JOURNAL *journal_parse(
	string_pipe(
		journal_system_string(
			JOURNAL_SELECT,
			JOURNAL_TABLE,
			journal_transaction_account_where() ),
	0 /* not fetch_check_number */,
	0 /* not fetch_memo */ );

/* Usage */
void journal_propagate(
	char *transaction_date_time,
	char *account_name );

/* Process */
JOURNAL *journal_prior(
	transaction_date_time,
	account_name );

LIST *journal_list_prior(
	journal_prior(),
	account_name );

LIST *journal_list_set_balance(
	LIST *journal_list_prior() /* in/out */,
	boolean account_accumulate_debit(
		account_name,
		0 /* not expect_lots */ ) );

void journal_list_update(
	journal_list_set_balance() );

/* Usage */
void journal_list_update(
	LIST *journal_list_set_balance() );

/* Process */
char *journal_update_system_string(
	char *JOURNAL_TABLE );

FILE *update_pipe =
	popen( journal_update_system_string(), “w” );

for JOURNAL *journal in journal_list_set_balance
{
	fprintf( update_pipe, journal->transaction_count );
	fprintf( update_pipe, journal->previous_balance );
	fprintf( update_pipe, journal->balance );
}

pclose( update_pipe );

/* Usage */
void journal_account_name_list_propagate(
	char *transaction_date_time,
	LIST *account_name_list );

/* Process */
for account_name in account_name_list
{
	journal_propagate(
		transaction_date_time,
		account_name );
}
```

JOURNAL (2)

```
/* Usage */
LIST *journal_system_list(
	char *journal_system_string(
		JOURNAL_SELECT,
		JOURNAL_TABLE,
		journal_transaction_where() ),
	boolean fetch_check_number,
	boolean fetch_memo );

/* Process */
LIST *list = list_new();
FILE *pipe = popen( journal_system_string(), “r” );

for input in string_input( pipe )
{
	list_set(
		list,
		journal_parse() );
}

/* Usage */
JOURNAL *journal_parse(
	char *input,
	boolean fetch_check_number,
	boolean fetch_memo );

JOURNAL *journal_new(
	char *full_name,
	char *street_address,
	char *transaction_date_time,
	char *account_name );

/* Attributes */
char *full_name;
char *street_address;
char *transaction_date_time;
char *account_name;
double previous_balance;
double debit_amount;
double credit_amount;
double balance;
int transaction_count;
int check_number;
char *memo;
boolean match_sum_taken;
double journal_amount;

if ( fetch_check_number )
{
	int check_number =
		atoi(
			journal_transaction_cell_fetch(
				full_name,
				street_address,
				transaction_date_time,
				“check_number” ) );
}
else
if ( fetch_memo )
{
	char *memo =
		journal_transaction_cell_fetch(
			full_name,
			street_address,
			transaction_date_time,
			“memo” );
}

/* Usage */
char *journal_transaction_cell_fetch(
	char *full_name,
	char *street_address,
	char *transaction_date_time,
	char *attribute_name );

/* Process */
char *transaction_primary_where(
	full_name,
	street_address,
	transaction_date_time );

char *journal_transaction_system_string(
	attribute_name,
	TRANSACTION_TABLE,
	transaction_primary_where() );

char *string_input(
	journal_transaction_system_string() );

/* Usage */
JOURNAL *journal_prior(
	char *transaction_date_time,
	char *account_name );

/* Process */
char *journal_max_prior_where(
	char *transaction_date_time,
	char *account_name );

char *journal_max_prior_transaction_date_time(
	char *journal_max_prior_where(),
	char *JOURNAL_TABLE );

JOURNAL *journal_account_fetch(
	journal_max_prior_transaction_date_time()
		/* transaction_date_time */,
	account_name );

/* Usage */
JOURNAL *journal_latest(
	char *transaction_date_time_closing,
	char *account_name );

/* Process */
char *journal_max_where(
	char *transaction_date_time_closing,
	char *account_name );

char *journal_max_transaction_date_time(
	char *journal_max_where(),
	char *JOURNAL_TABLE );

JOURNAL *journal_account_fetch(
	journal_max_transaction_date_time()
		/* transaction_date_time */,
	account_name );
```

JOURNAL (3)

```
/* Usage */
LIST *journal_account_journal_list(
   char *account_name,
   char *begin_transaction_date_time,
   char *end_transaction_date_time );

/* Process */
char *journal_account_where(
   char *account_name,
   char *begin_transaction_date_time,
   char *end_transaction_date_time );

LIST *journal_system_list(
   journal_system_string(
         JOURNAL_SELECT,
         JOURNAL_TABLE,
         journal_account_where() ),
   0 /* not fetch_check_number */,
   0 /* not fetch_memo */ );

/* Usage */
LIST *journal_POR_list(
   char *account_name );

/* Process */
char *account_primary_where(
   account_name );

char *journal_system_string(
   JOURNAL_SELECT,
   JOURNAL_TABLE,
   account_primary_where() );

LIST *journal_system_list(
   journal_system_string(),
   0 /* not fetch_check_number */,
   0 /* not fetch_memo */ );

/* Usage */
LIST *journal_list_prior(
   JOURNAL *journal_prior(),
   char *account_name );

/* Process */
if ( journal_prior )
{
   LIST *journal_list =
         journal_list_minimum(
               journal_prior->transaction_date_time
                     /* minimum_transaction_date_time */,
               account_name );
}
else
{
   LIST *journal_list =
         journal_POR_list(
               account_name );

   JOURNAL *first_journal = list_first( journal_list );
   first_journal->previous_balance = 0.0;
   first_journal->transaction_count = 1;
}

if ( journal_prior && list_length( journal_list ) == 1 )
{
   journal_list = (LIST *)0;
}

/* Usage */
JOURNAL *journal_debit_new(
      char *debit_account_name,
      double debit_amount );

/* Process */
JOURNAL *journal_calloc();

/* Usage */
JOURNAL *journal_credit_new(
      char *credit_account_name,
      double credit_amount );

/* Process */
JOURNAL *journal_calloc();

/* Usage */
void journal_list_insert(
   char *appaserver_error_filename,
   char *full_name,
   char *street_address,
   char *transaction_date_time,
   LIST *journal_list );

/* Process */
FILE *journal_insert_open(
   char *appaserver_error_filename,
   char *JOURNAL_INSERT,
   char *JOURNAL_TABLE );

void journal_insert(
   FILE *pipe,
   char *full_name,
   char *street_address,
   char *transaction_date_time,
   char *account_name,
   double debit_amount,
   double credit_amount );

/* Usage */
void journal_list_html_display(
   LIST *journal_list,
   char *transaction_memo );

/* Process */
void journal_list_pipe_display(
   FILE *output_pipe,
   char *transaction_memo,
   char *heading,
   LIST *journal_list );
```

JOURNAL (4)

```
/* Usage */
double journal_list_transaction_amount(
   LIST *journal_list );

/* Process */
static LIST *account_cash_name_list(
      ACCOUNT_TABLE,
      SUBCLASSIFICATION_CASH );

static LIST *account_current_liability_name_list(
      ACCOUNT_TABLE,
      SUBCLASSIFICATION_CURRENT_LIABILITY,
      ACCOUNT_UNCLEARED_CHECKS );

double cash_sum =
   journal_account_list_debit_sum(
         journal_list,
         account_cash_name_list() );

double current_liability_sum =
   journal_account_list_credit_sum(
         journal_list,
         account_current_liability_name_list() );

if ( cash_sum )
{
   return cash_sum;
}
else
if ( current_liability_sum )
{
   return current_liability_sum;
}
else
{
   return journal_debit_sum( journal_list );
}

/* Usage */
double journal_account_list_debit_sum(
   LIST *journal_list,
   LIST *account_name_list );

/* Process */
for JOURNAL *journal in journal_list
{
   if ( list_string_exists(
         journal->account_name,
         account_name_list ) )
   {
         if ( journal->debit_amount )
               sum += journal->debit_amount;
         else
               sum -= journal->credit_amount;
   }
}

/* Usage */
double journal_account_list_credit_sum(
   LIST *journal_list,
   LIST *account_name_list );

/* Process */
for JOURNAL *journal in journal_list
{
   if ( list_string_exists(
         journal->account_name,
         account_name_list ) )
   {
         if ( journal->credit_amount )
               sum += journal->credit_amount;
         else
               sum -= journal->debit_amount;
   }
}

/* Usage */
double journal_debit_sum(
   LIST *journal_list );

/* Process */
for JOURNAL *journal in journal_list
{
   if ( journal->debit_amount )
         sum += journal->debit_amount;
}

/* Private */
JOURNAL *journal_calloc( void );

/* Public */
char *journal_system_string(
   char *JOURNAL_SELECT,
   char *JOURNAL_TABLE,
   char *where );

char *journal_transaction_system_string(
   char *attribute_name,
   char *JOURNAL_TRANSACTION_TABLE,
   char *where );

JOURNAL *journal_list_latest(
   LIST *journal_list );

double journal_credit_debit_difference_sum(
   LIST *journal_list );

double journal_debit_credit_difference_sum(
   LIST *journal_list );

LIST *journal_account_distinct_entity_list(
   char *JOURNAL_TABLE,
   LIST *account_name_list );

LIST *journal_extract_account_name_list(
   LIST *journal_list );

LIST *journal_fetch_account_name_list(
   char *JOURNAL_TABLE,
   char *transaction_primary_where()
         /* where */ );

char *journal_delete_system_string(
   char *JOURNAL_TABLE,
   char *transaction_primary_where()
         /* where */ );

LIST *journal_binary_list(
   char *full_name,
   char *street_address,
   char *transaction_date_time,
   double transaction_amount,
   char *debit_account,
   char *credit_account );

char *journal_list_raw_display(
   LIST *journal_list );
```

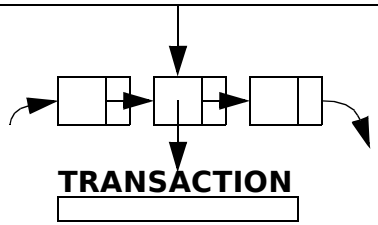
JOURNAL_PROPERTY

TRANSACTION_LIST

```
/* Usage */
TRANSACTION_LIST *transaction_list_fetch(
      char *where,
      boolean fetch_journal_ledger );

/* Process */
TRANSACTION_LIST *transaction_list_calloc();

char *transaction_system_string(
      TRANSACTION_SELECT,
      TRANSACTION_TABLE,
      where );

FILE *input_pipe =
      popen(
            transaction_system_string(),
            “r” );

LIST *list = list_new();

for input in string_input()
{
      list_set(
            list,
            transaction_parse(
                  input,
                  fetch_journal_ledger ) );
}

pclose( input_pipe );
```

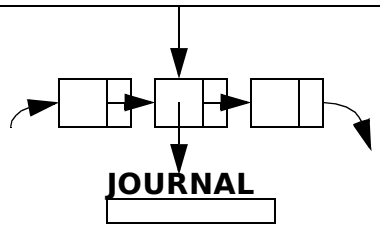
```
/* Usage */
TRANSACTION_LIST *transaction_list_new(
      LIST *list );

/* Process */
TRANSACTION_LIST *transaction_list_calloc();

transaction_list->list = list;

/* Usage */
void transaction_list_insert(
      TRANSACTION_LIST *transaction_list,
      char *appaserver_error_filename );

/* Process */
for TRANSACTION *transaction in transaction_list->list
{
      transaction->transaction_date_time =
            transaction_insert(
                  appaserver_error_filename,
                  transaction->full_name,
                  transaction->street_address,
                  transaction->transaction_date_time,
                  transaction->transaction_amount,
                  transaction->check_number,
                  transaction->memo,
                  TRANSACTION_LOCK_Y,
                  transaction->journal_list );
}
```

```
/* Usage */
void transaction_list_html_display(
      TRANSACTION_LIST *transaction_list );

/* Process */
for TRANSACTION *transaction in transaction_list->list
{
      journal_list_html_display(
            transaction->journal_list,
            transaction->memo ) );
}

/* Private */
TRANSACTION_LIST *transaction_list_calloc( void );

/* Public */
TRANSACTION *transaction_list_check_seek(
      int check_number,
      TRANSACTION_LIST *transaction_list );

TRANSACTION *transaction_list_entity_seek(
      char *full_name,
      char *street_address,
      char *transaction_date_time,
      TRANSACTION_LIST *transaction_list );

LIST *transaction_list_account_name_list(
      TRANSACTION_LIST *transaction_list );
```



TRANSACTION (1)

```
/* Usage */
TRANSACTION *transaction_fetch(
        char *full_name,
        char *street_address,
        char *transaction_date_time,
        boolean fetch_journal_ledger );

/* Process */
char *transaction_primary_where(
        full_name,
        street_address,
        transaction_date_time );

char *transaction_system_string(
        TRANSACTION_SELECT,
        TRANSACTION_TABLE,
        transaction_primary_where()
                /* where */ );

TRANSACTION *transaction_parse(
        char *string_pipe_fetch(
                transaction_system_string() )
                        /* input */,
        boolean fetch_journal_ledger );

TRANSACTION *transaction_new(
        full_name,
        street_address,
        transaction_date_time );

TRANSACTION *transaction_calloc();

/* Attributes */
char *full_name;
char *street_address;
char *transaction_date_time;
double transaction_amount;
char *memo;
int check_number;
boolean lock_transaction;
LIST *journal_list;

/* Usage
TRANSACTION *transaction_entity_new(
        ENTITY *entity,
        char *transaction_date_time,
        double transaction_amount,
        int check_number,
        char *memo,
        LIST *journal_list );

/* Process */
TRANSACTION *transaction_calloc();

/* Usage */
TRANSACTION *transaction_binary(
        char *full_name,
        char *street_address,
        char *transaction_date_time,
        double transaction_amount,
        char *memo,
        char *debit_account,
        char *credit_account );

/* Process */
TRANSACTION *transaction_new(
        full_name,
        street_address,
        transaction_date_time );

LIST *journal_list =
        journal_binary_list(
                full_name,
                street_address,
                transaction_date_time,
                transaction_amount,
                debit_account,
                credit_account );
```



TRANSACTION (2)

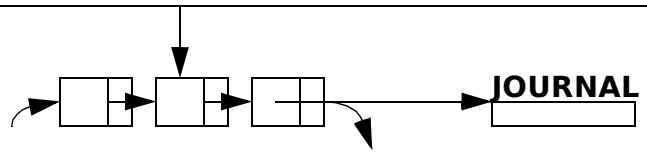
```
/* Usage */
/* Returns inserted transaction_date_time */
char *transaction_insert(
   char *appaserver_error_filename,
   char *full_name,
   char *street_address,
   char *transaction_date_time,
   double transaction_amount,
   int check_number,
   char *memo,
   char lock_transaction_yn,
   LIST *journal_list );

/* Process */
LIST *journal_extract_account_name_list(
   journal_list );

FILE *transaction_insert_pipe_open(
   char *appaserver_error_filename,
   char *TRANSACTION_SELECT,
   char *TRANSACTION_TABLE );

char *race_free_date_time =
   transaction_race_free_date_time(
         char *transaction_date_time );

void transaction_insert_pipe(
   FILE *transaction_insert_pipe_open(),
   char *full_name,
   char *street_address,
   char *race_free_date_time,
   double transaction_amount,
   char *transaction_check_number(
         int check_number ),
   char *transaction_memo(
         char *memo ),
   char *transaction_lock(
         char transaction_lock_yn ) );

pclose( transaction_insert_pipe_open() );

void journal_list_insert(
   appaserver_error_filename,
   full_name,
   street_address,
   race_free_date_time,
   journal_list );

void journal_account_name_list_propagate(
   race_free_date_time,
   journal_extract_account_name_list() );

/* Usage */
void transaction_delete(
   full_name,
   street_address,
   transaction_date_time );

/* Process */
char *transaction_primary_where(
   full_name,
   street_address,
   transaction_date_time );

char *transaction_delete_system_string(
   char *TRANSACTION_TABLE,
   char *transaction_primary_where() );

system( transaction_delete_system_string() );

LIST *journal_fetch_account_name_list(
   JOURNAL_TABLE,
   transaction_primary_where(
         full_name,
         street_address,
         transaction_date_time ) );

char *journal_delete_system_string(
   JOURNAL_TABLE,
   transaction_primary_where(
         full_name,
         street_address,
         transaction_date_time ) );

system( journal_delete_system_string() );

void journal_account_name_list_propagate(
   transaction_date_time,
   journal_fetch_account_name_list() );

/* Usage */
boolean transaction_closing_entry_exists(
   char *TRANSACTION_TABLE,
   char *PREDICTIVE_CLOSE_TIME,
   char *transaction_date );

/* Process */
char *transaction_date_time_where(
   TRANSACTION_DATE_TIME_COLUMN,
   transaction_date,
   PREDICTIVE_CLOSE_TIME );

char *transaction_closing_memo_where(
   char *TRANSACTION_CLOSING_ENTRY_MEMO );

char *transaction_closing_entry_where(
   char *transaction_date_time_where(),
   char *transaction_closing_memo_where() );

char *transaction_closing_entry_system_string(
   char *TRANSACTION_TABLE,
   char *transaction_closing_entry_where() );

( atoi(
   pipe2string(
         transaction_closing_entry_system_string() )
 == 1 );

/* Usage */
void transaction_update(
   double transaction_amount,
   char *transaction_primary_where() );

/* Process */
char *transaction_set_clause(
   char *TRANSACTION_AMOUNT_COLUMN,
   double transaction_amount );

char *transaction_update_statement(
   char *TRANSACTION_TABLE,
   transaction_set_clause(),
   transaction_primary_where );

system( transaction_update_statement );
```

TRANSACTION (3)

```
/* Usage */
char *transaction_date_time_where(
   char *TRANSACTION_DATE_TIME_COLUMN,
   char *transaction_date,
   char *transaction_time );

/* Process */
char *transaction_date_time(
   char *transaction_date,
   char *transaction_time );

/* Private */
TRANSACTION *transaction_calloc( void );

/* Public */
char *transaction_primary_where(
      char *full_name,
      char *street_address,
      char *transaction_date_time );

char *transaction_system_string(
      char *TRANSACTION_SELECT,
      char *TRANSACTION_TABLE,
      char *where );

char *transaction_date_time_closing(
      char *PREDICTIVE_CLOSE_TIME,
      char *PREDICTIVE_PRECLOSE_TIME,
      char *transaction_date,
      boolean preclose_time,
      boolean transaction_closing_entry_exists() );

char *transaction_raw_display(
      TRANSACTION *transaction );

boolean transaction_date_time_changed(
      char *preupdate_transaction_date_time );

boolean transaction_date_time_exists(
      char *TRANSACTION_TABLE,
      char *TRANSACTION_DATE_TIME_COLUMN,
      char *transaction_date_time );
```

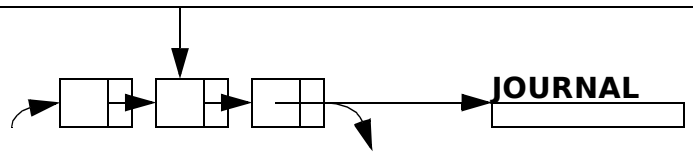
TRANSACTION_PROGRAM

```
/* Usage */
TRANSACTION *transaction_program_fetch(
        full_name,
        street_address,
        transaction_date_time,
        char *program_name,
        boolean fetch_journal_ledger );

/* Process */
```



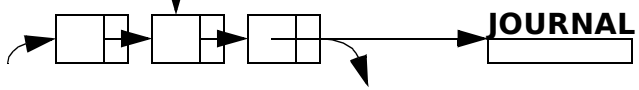
TRANSACTION_FUND

```
/* Usage */
TRANSACTION *transaction_program_fetch(
        full_name,
        street_address,
        transaction_date_time,
        char *fund_name,
        boolean fetch_journal_ledger );

/* Process */
```



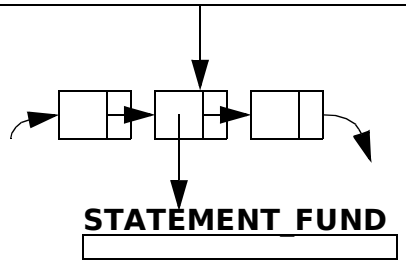
TRANSACTION_PROPERTY

```
/* Usage */
TRANSACTION *transaction_property_fetch(
        full_name,
        street_address,
        transaction_date_time,
        char *property_street_address,
        boolean fetch_journal_ledger );

/* Process */
```



STATEMENT (1)

```
/* Usage */
STATEMENT *statement_new(
    char *application_name,
    char *session,
    char *login_name,
    char *role_name,
    char *process_name,
    boolean exists_logo_filename,
    LIST *filter_element_name_list,
    char *as_of_date,
    int prior_year_count,
    LIST *fund_name_list,
    char *subclassification_option_string,
    char *fund_aggregation_string,
    char *output_medium_string );

STATEMENT *statement_steady_state(
    char *application_name,
    char *session,
    char *login_name,
    char *role_name,
    char *process_name,
    boolean exists_logo_filename,
    LIST *filter_element_name_list,
    char *as_of_date,
    int prior_year_count,
    LIST *fund_name_list,
    char *subclassification_option_string,
    char *fund_aggregation_string,
    char *output_medium_string,
    boolean with_postclose,
    STATEMENT *statement );

enum subclassification_option
    statement_subclassification_option(
        subclassification_option_string );

enum fund_aggregation
    statement_fund_aggregation(
        fund_name_list,
        fund_aggregation_string );

enum output_medium
    statement_output_medium(
        output_medium_string );

LIST *statement_fund_name_list( void );

char *statement_begin_date_string(
    char *as_of_date );

char *statement_title(
    char *application_name,
    boolean exists_logo_filename,
    char *process_name );

char *statement_subtitle(
    statement_begin_date_string(),
    char *as_of_date,
    enum fund_aggregation );

LIST *statement_fund_list(
    application_name,
    session,
    login_name,
    role_name,
    filter_element_name_list,
    statement_begin_date_string(),
    as_of_date,
    prior_year_count,
    fund_name_list,
    char *progam_name,
    statement_subclassification_option(),
    statement_fund_aggregation(),
    boolean with_postclose );

boolean statement_exists_closing_entry(
    char *as_of_date );

void statement_account_delta_list(
    char *account_name,
    LIST *prior_year_list );

void statement_html_account_delta_list(
    char *account_name,
    LIST *prior_year_list );

void statement_subclassification_delta_list(
    char *subclassification_name,
    LIST *prior_year_list );

void statement_html_subclassification_delta_list(
    char *subclassification_name,
    LIST *prior_year_list );

void statement_PDF_account_delta_list(
    char *account_name,
    LIST *prior_year_list );

void statement_PDF_subclassification_delta_list(
    char *subclassification_name,
    LIST *prior_year_list );

int statement_delta_prior(
    double prior_total,
    double total );
```



statement_subclassification_option

subclassification_aggregate,
subclassification_display,
subclassification_omit

statement_fund_aggregation

single_fund,
sequential,
consolidated,
no_fund

statement_output_medium

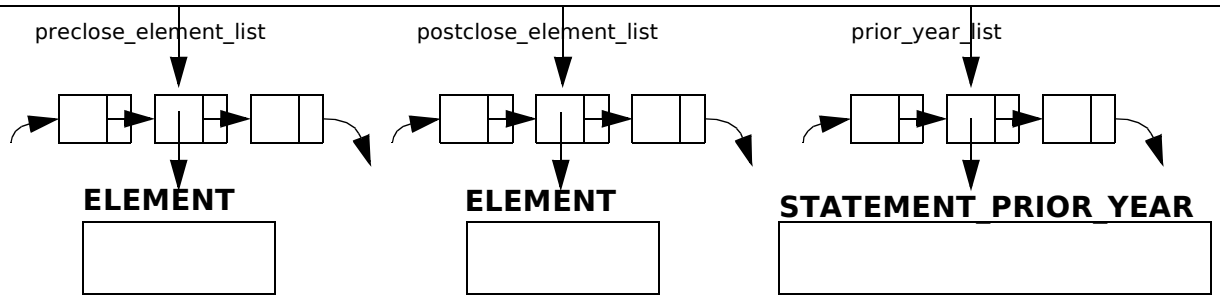
output_table,
output_PDF
output_spreadsheet,
output_stdout

STATEMENT (2)

```
/* Process */
SUBCLASSIFICATION *statement_net_income_subclassification(
        double *equity_element_current_balance,
        int *equity_element_percent_of_asset,
        SUBCLASSIFICATION *last_subclassification,
        EQUITY_ELEMENT *equity_element,
        char *fund_name,
        char *transaction_date_time_closing,
        double asset_element_current_balance,
        boolean is_financial_position );

ELEMENT *statement_equity_liability_element(
        double equity_element_current_balance,
        double asset_element_current_balance,
        double liability_element_current_balance );
```

STATEMENT_FUND (1)

```
/* Usage */
STATEMENT_FUND *statement_fund_fetch(
    char *application_name,
    char *session,
    char *login_name,
    char *role_name,
    LIST *filter_element_name_list,
    char *begin_date_string,
    char *as_of_date,
    int prior_year_count,
    char *fund_name,
    enum subclassification_option,
    boolean with_postclose );

/* Process */
char *transaction_date_time_nominal =
    transaction_date_time_closing(
        as_of_date,
        1 /* preclose_time_boolean */,
        transaction_closing_entry_exists(
            as_of_date ) );

char *transaction_date_time_fixed =
    transaction_date_time_closing(
        as_of_date,
        0 /* not preclose_time_boolean */,
        transaction_closing_entry_exists(
            as_of_date ) );

LIST *statement_fund_element_list(
    char *application_name,
    char *session,
    char *login_name,
    char *role_name,
    LIST *filter_element_name_list,
    char *begin_date_string,
    char *transaction_date_time_nominal,
    char *transaction_date_time_fixed,
    char *fund_name,
    enum subclassification_option );

LIST *preclose_element_list =
    statement_fund_element_list(
        application_name,
        session,
        login_name,
        role_name,
        filter_element_name_list,
        begin_date_string,
        transaction_date_time_nominal,
        transaction_date_time_nominal,
        fund_name,
        subclassification_option );

LIST *postclose_element_list =
    statement_fund_element_list(
        application_name,
        session,
        login_name,
        role_name,
        filter_element_name_list,
        begin_date_string,
        transaction_date_time_fixed,
        transaction_date_time_fixed,
        fund_name,
        subclassification_option );

LIST *prior_year_list;

for years_ago = 1 to prior_year_count
{
    list_set(
        prior_year_list,
        statement_prior_year_new(
            application_name,
            session,
            login_name,
            role_name,
            filter_element_name_list,
            as_of_date,
            years_ago,
            fund_name,
            subclassification_option ) );
}

void statement_fund_list_steady_state(
    LIST *statement_fund_list,
    char *title,
    char *subtitle,
    boolean exists_logo_filename );

STATEMENT_FUND *statement_fund_steady_state(
    LIST *preclose_element_list,
    LIST *postclose_element_list,
    char *title,
    char *subtitle,
    boolean exists_logo_filename,
    STATEMENT_FUND *statement_fund );

void statement_fund_list_prior_year_list_set(
    LIST *statement_fund_list,
    int prior_year_count );

LIST *statement_fund_steady_state_prior_year_list(
    int prior_year_count,
    LIST *preclose_element_list,
    STATEMENT_FUND *statement_fund );

STATEMENT_FUND *statement_fund_seek(
    char *fund_name,
    LIST *statement_fund_list );
```

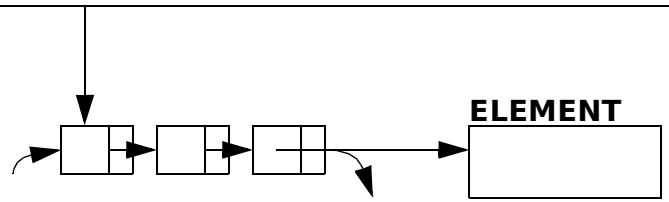


STATEMENT_FUND (2)

```
/* Process */
double preclose_debit_total =
    element_list_debit_total(
        preclose_element_list );

double preclose_credit_total =
    element_list_credit_total(
        preclose_element_list );

double postclose_debit_total =
    element_list_debit_total(
        postclose_element_list );

double postclose_credit_total =
    element_list_credit_total(
        postclose_element_list );

boolean statement_fund_exists_postclose(
    LIST *statement_fund_list );

void statement_fund_list_net_income_set(
    LIST *statement_fund_list );

void statement_fund_list_net_income_fetch(
    LIST *statement_fund_list );

double statement_fund_net_income_fetch(

boolean  statement_fund_exists_prior_year(
    LIST *statement_fund_list );

double statement_fund_total(
    LIST *subclassification_list,
    LIST *account_list );

double revenue_total =
    element_seek(
        ELEMENT_REVENUE,
        preclose_element_list )->
            element_current_balance() );

double expense_total =
    element_seek(
        ELEMENT_EXPENSE,1
        preclose_element_list )->
            element_current_balance() );

double gain_total =
    element_seek(
        ELEMENT_GAIN,
        preclose_element_list )->
            element_current_balance() );

double loss_total =
    element_seek(
        ELEMENT_LOSS,
        preclose_element_list )->
            element_current_balance() );

double net_income =
    statement_fund_net_income(
        revenue_total,
        expense_total,
        gain_total,
        loss_total );

int statement_fund_net_income_percent(
    statement_fund_net_income(),
    revenue_total );

char *statement_fund_caption(
    char *statement_title(),
    char *statement_subtitle(),
    char *statement_fund->fund_name,
    boolean exists_logo_filename );

void statement_fund_list_equity_set(
    LIST *statement_fund_list,
    boolean is_financial_position,
    enum subclassification_option );

void statement_fund_display_equity(
    double *element_current_balance,
    int *percent_of_asset,
    element->subclassification_list /* out */,
    element->equity_element /* in/out */,
    double net_income,
    double asset_current_balance,
    boolean is_financial_position );

void statement_fund_omit_equity(
    double *element_current_balance,
    int *percent_of_asset,
    LIST *account_list /* out */,
    element->equity_element /* in/out */,
    double net_income,
    double asset_current_balance,
    boolean is_financial_position );

void statement_fund_aggregate_equity(
    double *element_current_balance,
    int *percent_of_asset,
    LIST *subclassification_list /* out */,
    element->equity_element /* in/out */,
    double net_income,
    double asset_current_balance,
    boolean is_financial_position );
```

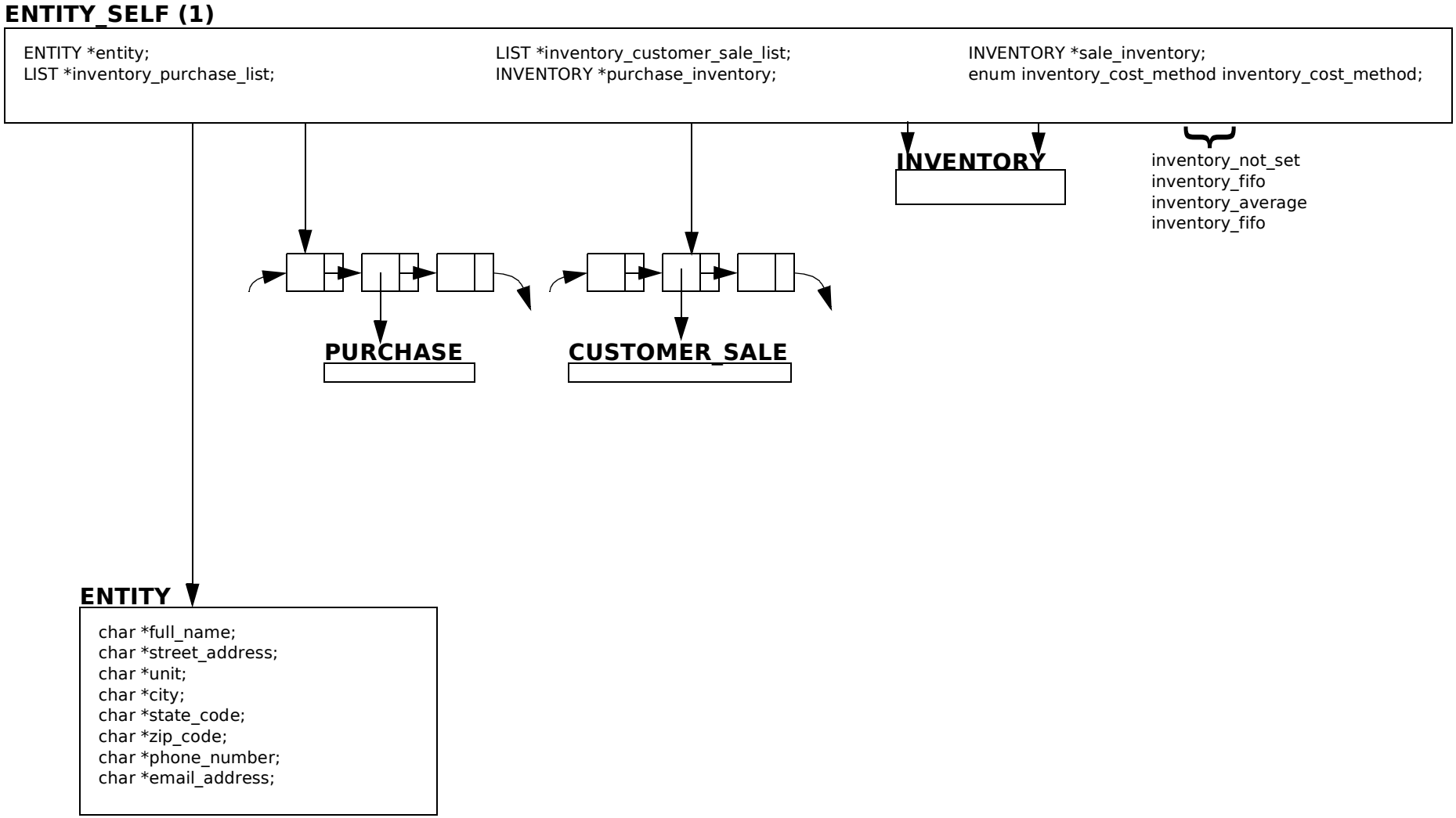
STATEMENT_PRIOR_YEAR

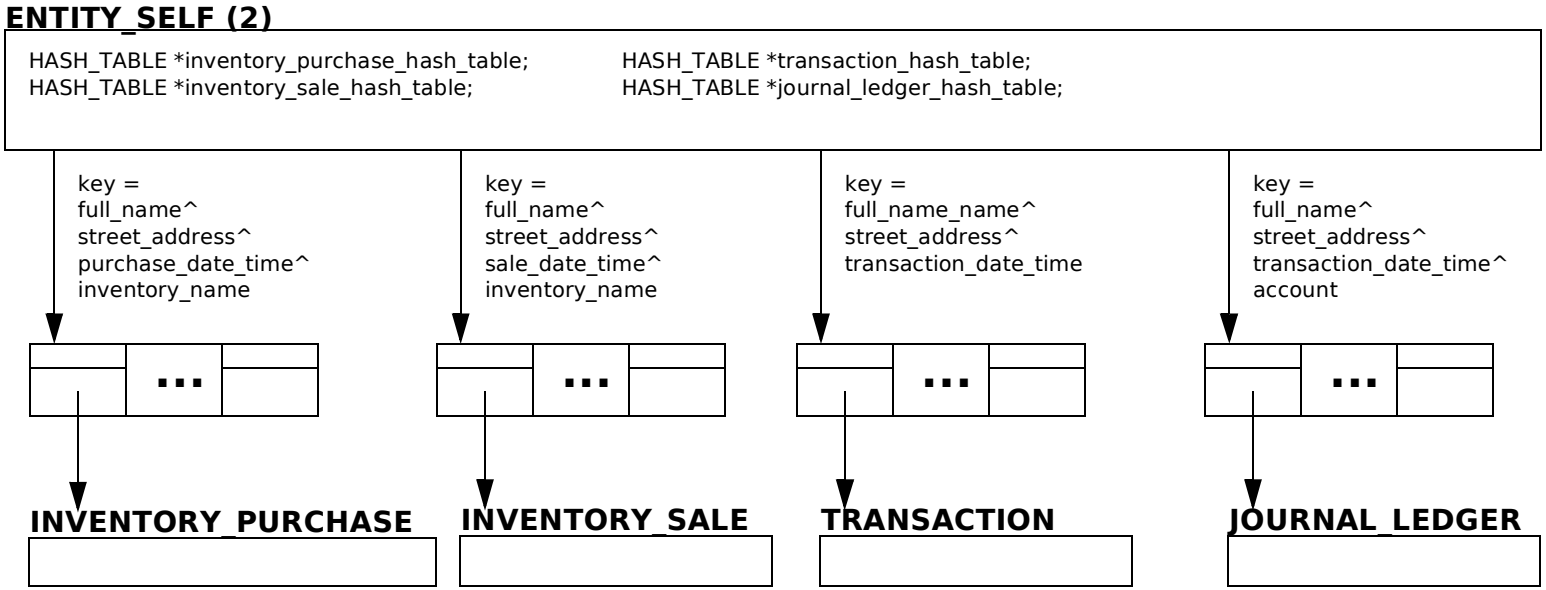
```
/* Usage */
STATEMENT_PRIOR_YEAR *
    statement_prior_year_new(
        char *application_name,
        char *session,
        char *login_name,
        char *role_name,
        LIST *filter_element_name_list,
        char *as_of_date,
        int years_ago,
        char *fund_name,
        enum subclassification_option );

/* Attributes */
char *application_name;
char *session;
char *login_name;
char *role_name;
LIST *filter_element_name_list;
char *as_of_date;
int years_ago;
char *fund_name;
enum subclassification_option subclassification_option;

/* Process */
char *prior_date_string =
    strdup(
        date_yyyy_mm_dd_display(
            date_subtract_year(
                date_yyyy_mm_dd_new(
                    as_of_date ),
                years_ago ) ) );

char *transaction_date_time_nominal =
    transaction_date_time_closing(
        prior_date_string,
        1 /* preclose_time_boolean */,
        transaction_closing_entry_exists(
            prior_date_string ) );

char *transaction_date_time_fixed =
    transaction_date_time_closing(
        prior_date_string,
        0 /* not preclose_time_boolean */,
        transaction_closing_entry_exists(
            prior_date_string ) );

char *begin_date_string =
    statement_begin_date_string(
        prior_date_string );

LIST *prior_year_element_list =
    statement_fund_element_list(
        application_name,
        session,
        login_name,
        role_name,
        filter_element_name_list,
        begin_date_string,
        transaction_date_time_nominal,
        transaction_date_time_fixed,
        fund_name,
        subclassification_option );

LIST *statement_prior_year_heading_list(
    LIST *prior_year_list );

LIST *statement_prior_year_delta_list(
    char *account_name,
    LIST *prior_year_list );

void statement_prior_year_list_net_income_set(
    LIST *prior_year_list,
    statement_fund_net_income() );

void statement_prior_year_net_income_set(
    STATEMENT_PRIOR_YEAR
        *statement_prior_year,
        statement_fund_net_income );

double revenue_total =
    element_seek(
        ELEMENT_REVENUE,
        prior_year_element_list )->
            element_current_balance() );

double expense_total =
    element_seek(
        ELEMENT_EXPENSE,
        prior_year_element_list )->
            element_current_balance() );

double gain_total =
    element_seek(
        ELEMENT_GAIN,
        prior_year_element_list )->
            element_current_balance() );

double loss_total =
    element_seek(
        ELEMENT_LOSS,
        prior_year_element_list )->
            element_current_balance() );

double statement_prior_year_net_income =
    statement_fund_net_income(
        revenue_total,
        expense_total,
        gain_total,
        loss_total );

int delta_prior =
    statement_delta_prior(
        statement_prior_year_net_income,
            statement_fund_net_income );
```



ENTITY

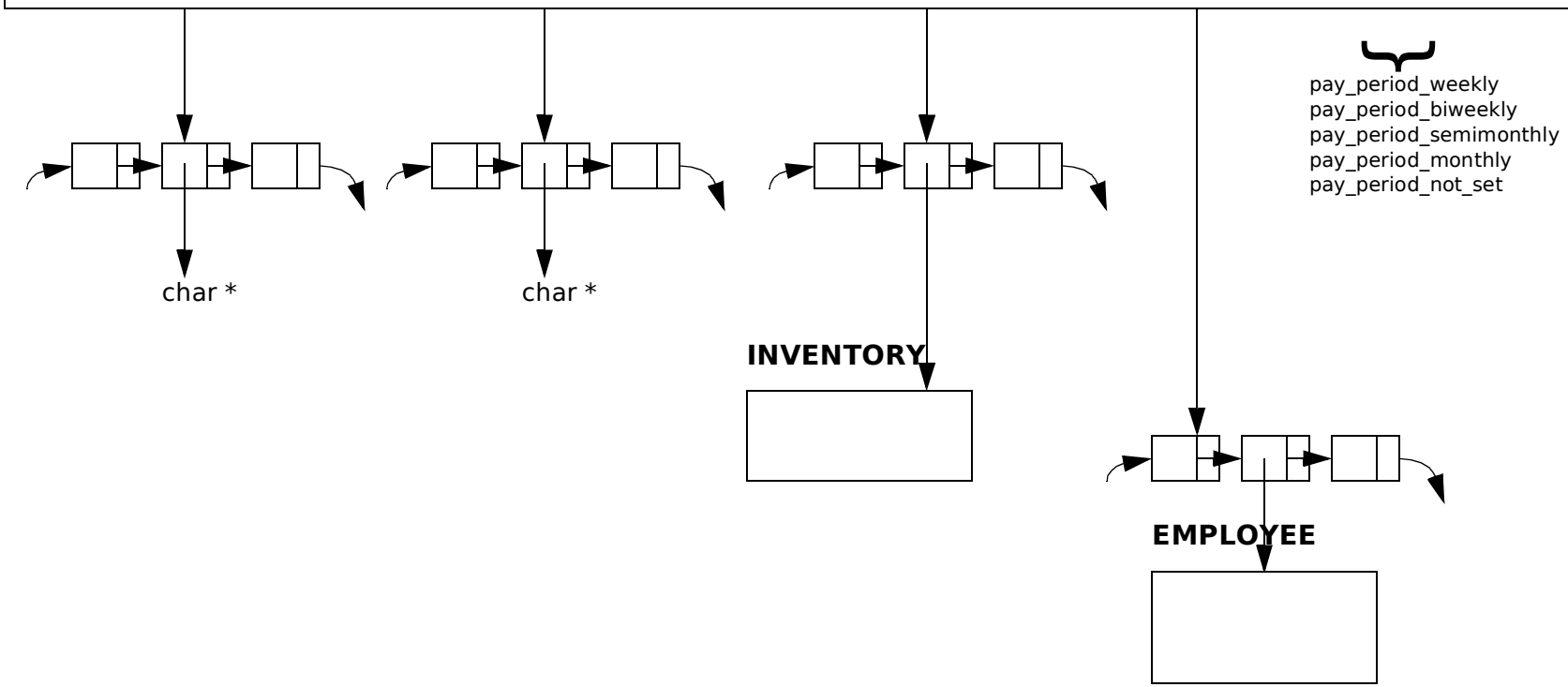
```
/* Usage */
ENTITY *entity_fetch(
	char *full_name,
	char *street_address );

/* Process */
char *entity_primary_where(
	char *full_name,
	char *street_address );

char *entity_system_string(
	char *ENTITY_SELECT,
	char *ENTITY_TABLE,
	char *entity_primary_where );

ENTITY *entity_parse(
	char *string_input() );

ENTITY *entity_new(
	char *strdup( full_name ),
	char *strdup( street_address ) );

ENTITY *entity_calloc( void );

/* Attributes */
char *full_name;
char *street_address;
char *unit;
char *city;
char *state_code;
char *zip_code;
char *phone_number;
char *email_address;

/* Public */
LIST *entity_full_street_list(
	LIST *full_name_list,
	LIST *street_address_list );
```


ENTITY_SELF (1)
ENTITY *entity;
LIST *inventory_purchase_list;
LIST *inventory_customer_sale_list;
INVENTORY *purchase_inventory;
INVENTORY *sale_inventory;
enum inventory_cost_method inventory_cost_method;
INVENTORY
inventory_not_set
inventory_fifo
inventory_average
inventory_fifo
PURCHASE
CUSTOMER_SALE
ENTITY
char *full_name;
char *street_address;
char *unit;
char *city;
char *state_code;
char *zip_code;
char *phone_number;
char *email_address;


ENTITY_SELF (2)
HASH_TABLE *inventory_purchase_hash_table;
HASH_TABLE *inventory_sale_hash_table;
HASH_TABLE *transaction_hash_table;
HASH_TABLE *journal_ledger_hash_table;
key =
full_name^
street_address^
purchase_date_time^
inventory_name
key =
full_name^
street_address^
sale_date_time^
inventory_name
key =
full_name_name^
street_address^
transaction_date_time
key =
full_name^
street_address^
transaction_date_time^
account
...
...
...
...
INVENTORY_PURCHASE
INVENTORY_SALE
TRANSACTION
JOURNAL_LEDGER

ENTITY_SELF (3)

```
LIST *inventory_purchase_name_list;
LIST *inventory_sale_name_list;
LIST *inventory_list;
LIST *employee_list;
enum payroll_pay_period payroll_pay_period;
```



ENTITY_SELF (4)

```
char *payroll_beginning_day
double social_security_combined_tax_rate
int social_security_payroll_ceiling
double medicare_combined_tax_rate
double medicare_additional_withholding_rate
int medicare_additional_gross_pay_floor
double federal_withholding_allowance_period_value
double federal_nonresident_withholding_income_premium
double state_withholding_allowance_period_value
double state_itemized_allowance_period_value
int federal_unemployment_wage_base
double federal_unemployment_tax_standard_rate
double federal_unemployment_threshold_rate
double federal_unemployment_tax_minimum_rate
int state_unemployment_wage_base
double state_unemployment_tax_rate
double state_sales_tax_rate;

char *entity_self_paypal_cash_account_name();
```

SALE (1)

```
ENTITY *customer_entity =
      entity_fetch(
            full_name,
            street_address );

char *sale_date_time;
char *completed_date_time;
double shipping_revenue;
enum title_passage_rule title_passage_rule;

/* Returns sale_steady_state() */
SALE *sale_fetch(
      full_name,
      street_address,
      sale_date_time );

char *sale_primary_where(
      full_name,
      street_address,
      sale_date_time );

double sale_gross_revenue(
      inventory_sale_total(),
      fixed_service_sale_total(),
      hourly_service_sale_total() );

double sale_sales_tax(
      inventory_sale_list(),
      entity_self_sales_tax_rate() );

double sale_invoice_amount(
      sale_gross_revenue(),
      sale_sales_tax(),
      shipping_charge );

double Sale_amount_due(
      sale_invoice_amount(),
      customer_payment_total() )
{
      return A - B;
}

TRANSACTION *sale_transaction(
      full_name,
      street_address,
      completed_date_time,
      invoice_amount,
      gross_revenue,
      sale_tax,
      shipping_charge,
      account_receivable(),
      account_revenue(),
      account_shipping_revenue(),
      account_sales_tax_payable() );

void sale_update(
      inventory_sale_total(),
      fixed_service_sale_total(),
      hourly_service_sale_total(),
      sale_gross_revenue()
      sale_sales_tax(),
      sale_invoice_amount(),
      payment_total(),
      sale_amount_due(),
      sale_transaction()->
            transaction_date_time,
      full_name,
      street_address,
      sale_date_time );

SALE *sale_steady_state(
      full_name,
      street_address,
      sale_date_time,
      inventory_sale_list(),
      fixed_service_sale_list(),
      hourly_service_sale_list(),
      customer_payment_list(),
      shipping_charge,
      entity_self_sales_tax_rate(),
      completed_date_time,
      account_receivable(),
      account_revenue(),
      account_shipping_revenue(),
      account_sales_tax_payable() );
```

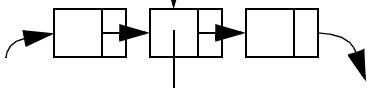
TRANSACTION

ENTITY

FOB_shipping
FOB_destination
title_passage_rule_null

SALE (2)

```
LIST *customer_payment_list(
      full_name,
      street_address,
      sale_date_time );
```

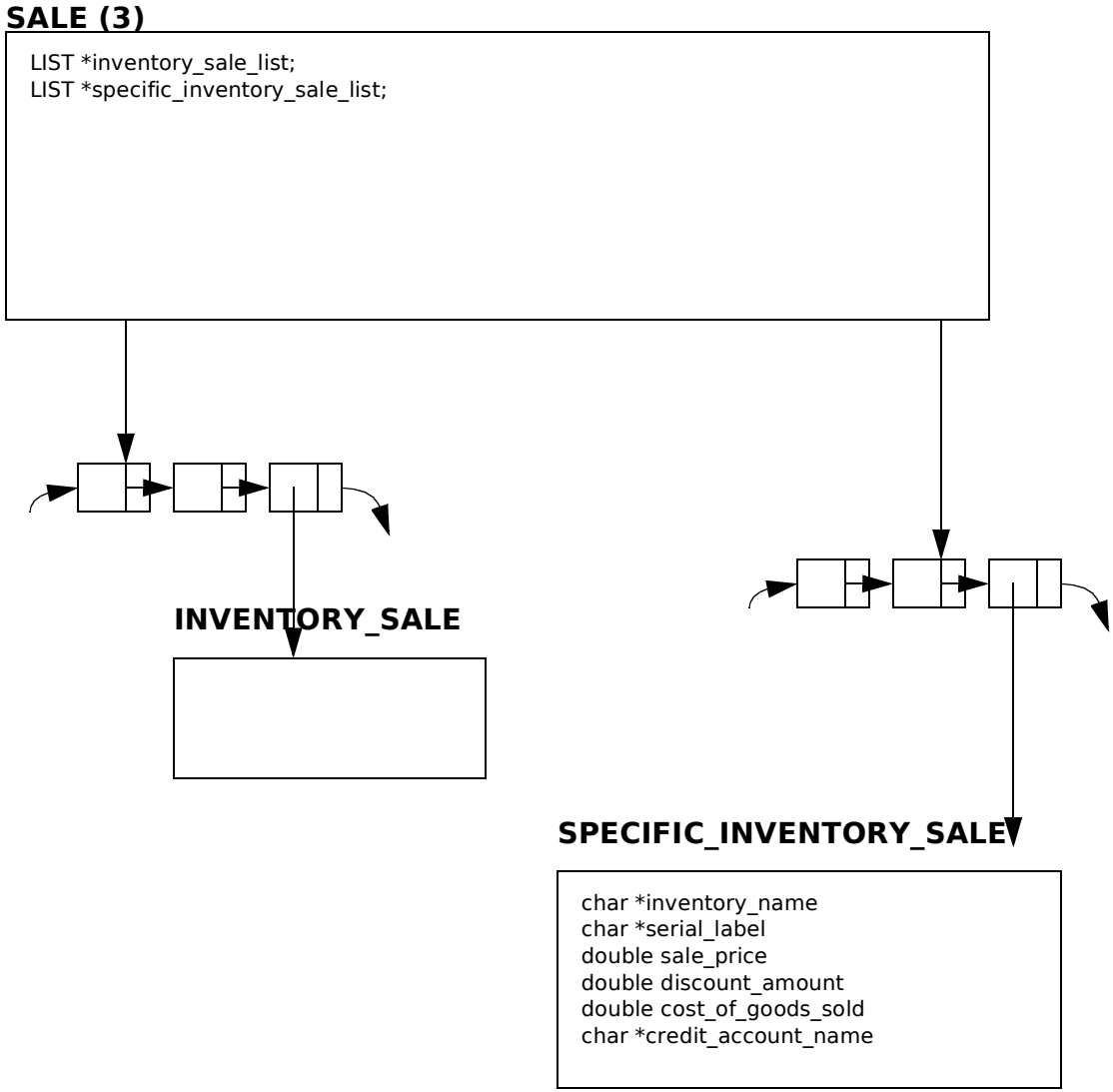


CUSTOMER_PAYMENT

```
ENTITY *customer_entity =
      entity_fetch(
            full_name,
            street_address );

char *sale_date_time;
char *payment_date_time;
double payment_amount;
int check_number;

double customer_payment_total(
      customer_payment_list() )
{
      return sum( A->payment_amount );
}

CUSTOMER_PAYMENT *customer_payment_new(
      full_name,
      street_address,
      sale_date_time,
      payment_date_time );

char *customer_payment_select(
      void );

char *customer_payment_sys_string(
      where );

LIST *customer_payment_system_list(
      sys_string );

CUSTOMER_PAYMENT *customer_payment_parse(
      input );

TRANSACTION *customer_payment_transaction(
      full_name,
      street_address,
      payment_date_time,
      payment_amount,
      account_cash(),
      account_receivable() );
```

ENTITY

TRANSACTION


SALE (3)
LIST *inventory_sale_list;
LIST *specific_inventory_sale_list;
INVENTORY_SALE
SPECIFIC_INVENTORY_SALE
char *inventory_name
char *serial_label
double sale_price
double discount_amount
double cost_of_goods_sold
char *credit_account_name

HOURLY_SERVICE_SALE

```
ENTITY *customer_entity =
    entity_new(
        full_name,
        street_address );
char *sale_date_time;
char *service_name;
char *service_description;
double hourly_rate;
double discount_amount;
int estimated_hours;

char *hourly_service_sale_revenue_account_name(
    service_name,
    service_description );

LIST *hourly_service_work_list =
    work_hourly_list(
        full_name,
        street_address,
        sale_date_time,
        service_name,
        service_description );

double hourly_service_sale_work_hours =
    work_list_hours(
        hourly_service_work_list() );

double hourly_service_sale_net_revenue(
    hourly_rate,
    hourly_service_sale_work_hours,
    discount_amount )
{
    return (A x B) - C;
}
```

SALE (4)

```
LIST *    fixed_service_sale_list
LIST *    hourly_service_sale_list
```

FIXED_SERVICE_SALE

```
ENTITY *customer_entity =
    entity_new(
        full_name,
        street_address );
char *sale_date_time;
char *service_name;
double fixed_price;
double discount_amount;
int estimated_hours;
char *fixed_service_sale_revenue_accouint_name(
    service_name );

LIST *fixed_service_sale_work_list =
    work_fixed_list(
        full_name,
        street_address,
        sale_date_time,
        service_name );

double fixed_service_sale_work_hours =
    work_list_hours(
        fixed_service_sale_work_list() );

double fixed_service_sale_net_revenue(
    fixed_price,
    discount_amount )
{
    return A - B;
}
```

WORK

```
ENTITY *customer_entity = entity_new();
char *sale_date_time;
char *service_name;
char *service_description;
char *begin_date_time;
char *end_date_time;
char *login_name;
char *activity_name;

double work_hours(
    begin_date_time,
    end_date_time );

double work_list_hours(
    work_list );
```

PURCHASE (1)

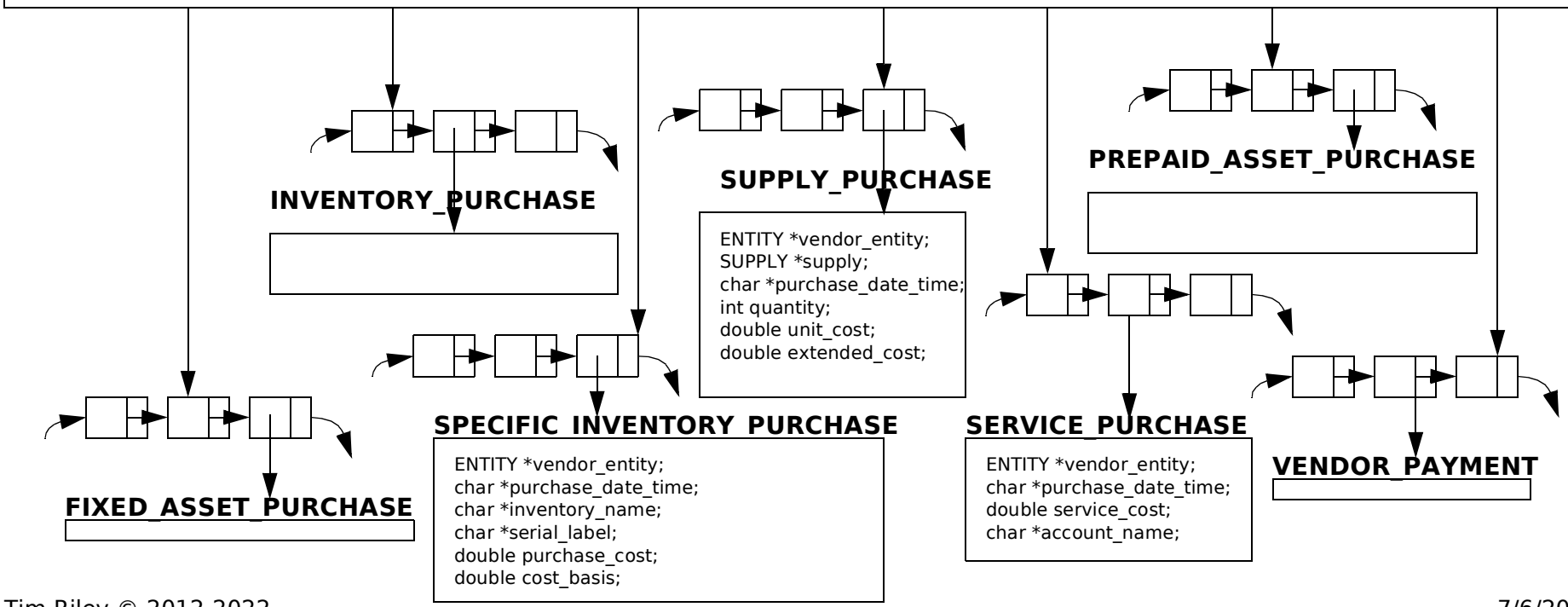
```
/* Usage */
PURCHASE *purchase_fetch(
    char *full_name,
    char *street_address,
    char *purchase_date_time );

/* Input */
ENTITY *vendor_entity =
    entity_new(
        full_name,
        street_address );

char *purchase_date_time;
double sales_tax;
double freight_in;
char *title_passage_rule_string;
char *shipped_date;
char *arrived_date_time;
char *program_name;
char *property_street_address;

/* Process */
enum title_passage_rule title_passage_rule =
    predictive_title_passage_rule_resolve(
        title_passage_rule_string );

char *purchase_primary_where(
    char *full_name,
    char *street_address,
    char *purchase_date_time );

char *purchase_system_string(
    char *where,
    char *order );

LIST *purchase_system_list(
    purchase_system_string() );

PURCHASE *purchase_parse(
    char *input );

LIST *fixed_asset_purchase_list =
    fixed_asset_purchase_list_fetch(
        vendor_entity->full_name,
        vendor_entity->street_address,
        purchase_date_time );

LIST *vendor_payment_list =
    vendor_payment_list_fetch(
        full_name,
        street_address,
        purchase_date_time );

PURCHASE *purchase_steady_state(
    sales_tax,
    freight_in,
    arrived_date,
    fixed_asset_purchase_list,
    inventory_purchase_list,
    specific_inventory_purchase_list,
    supply_purchase_list,
    prepaid_asset_purchase_list,
    inventory_purchase_return_list,
    vendor_payment_list,
    purchase );

double fixed_asset_purchase_total =
    fixed_asset_purchase_total(
        fixed_asset_purchase_list );

double inventory_purchase_total =
    inventory_purchase_total(
        inventory_purchase_list );

double specific_inventory_purchase_total =
    specific_inventory_purchase_total(
        specific_inventory_purchase_list );

double supply_purchase_total =
    supply_purchase_total(
        supply_purchase_list );

double prepaid_asset_purchase_total =
    prepaid_asset_purchase_total(
        prepaid_asset_purchase_list );

double inventory_purchase_return_total =
    inventory_purchase_return_total(
        inventory_purchase_return_list );

double vendor_payment_total =
    vendor_payment_total(
        vendor_payment_list );

double purchase_invoice_amount(
    fixed_asset_purchase_total,
    inventory_purchase_total,
    specific_inventory_purchase_total,
    supply_purchase_total,
    prepaid_asset_purchase_total,
    sales_tax,
    freight_in );

double purchase_amount_due(
        purchase_invoice_amount,
        vendor_payment_total )
{
    return A - B;
}

double purchase_asset_cost_basis(
        asset_cost,
        sales_tax,
        freight_in,
        fixed_asset_purchase_total(),
        inventory_purchase_total(),
        specific_inventory_purchase_total(),
        supply_expense_purchase_total(),
        prepaid_asset_purchase_total() );
```

ENTITY

FOB_shipping
FOB_destination
title_passage_rule_null

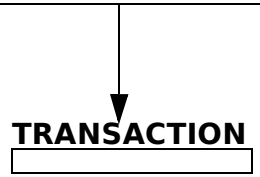
PURCHASE (2)

```
TRANSACTION *purchase_transaction =
    purchase_fixed_asset_transaction(
        full_name,
        street_address,
        arrived_date_time,
        purchase_invoice_amount,
        sales_tax,
        freight_in,
        fixed_asset_purchase_list(),
        account_payable() );

/* Returns true transaction_date_time */
char *purchase_transaction_refresh(
    purchase_transaction->
        transaction_amount,
    purchase_transaction->
        journal_list,
    purchase_transaction->memo,
    full_name,
    street_address,
    purchase_transaction->
        transaction_date_time );

void purchase_update(
    fixed_asset_purchase_total,
    purchase_invoice_amount,
    vendor_payment_total,
    purchase_amount_due,
    purchase_transaction->
        transaction_date_time,
    full_name,
    street_address,
    purchase_date_time );

LIST *inventory_purchase_list
LIST *specific_inventory_purchase_list
LIST *service_purchase_list
LIST *supply_purchase_list
LIST *prepaid_asset_purchase_list;
char *property_street_address;
```



VENDOR_PAYMENT

```
/* Usage */
VENDOR_PAYMENT *vendor_payment_fetch(
	char *full_name,
	char *street_address,
	char *purchase_date_time,
	char *payment_date_time );

LIST *vendor_payment_list_fetch(
	full_name,
	street_address,
	purchase_date_time );

/* Input */
ENTITY *vendor_entity;
char *purchase_date_time;
char *payment_date_time;
double payment_amount;
int check_number;

/* Process */
VENDOR_PAYMENT *vendor_payment_new(
	full_name,
	street_address,
	purchase_date_time,
	payment_date_time );

char *vendor_payment_primary_where(
	full_name,
	street_address,
	purchase_date_time,
	payment_date_time );

char *vendor_payment_system_string(
	purchase_primary_where() );

LIST *vendor_payment_system_list(
	vendor_payment_system_string() );

VENDOR_PAYMENT *vendor_payment_parse(
	input );

double vendor_payment_total(
	LIST *vendor_payment_system_list() )
{
	return sum(
			A->
			vendor_payment->
			payment_amount );
}

TRANSACTION *vendor_payment_transaction(
	full_name,
	street_address,
	payment_date_time,
	payment_amount,
	check_number,
	account_payable(),
	account_cash(),
	account_uncleared_checks() );

LIST *vendor_payment_journal_list(
	payment_amount,
	check_number,
	account_payable(),
	account_cash(),
	account_uncleared_checks() );

void vendor_payment_insert(
	full_name,
	street_address,
	purchase_date_time,
	payment_date_time,
	payment_amount,
	check_number );

VENDOR_PAYMENT *vendor_payment_seek(
	vendor_payment_list,
	payment_date_time );

char *vendor_payment_update_system_string(
	void );
```



FIXED_ASSET_PURCHASE

```
/* Usage */
FIXED_ASSET_PURCHASE *fixed_asset_purchase_fetch(
        char *asset_name,
        char *serial_label,
        boolean fetch_last_depreciation,
        boolean fetch_last_recovery );

LIST *fixed_asset_purchase_list_fetch(
        char *where,
        boolean fetch_last_depreciation,
        boolean fetch_last_recovery );

/* Input */
FIXED_ASSET *fixed_asset;
char *serial_label;
ENTITY *vendor_entity;
char *purchase_date_time;
char *service_placement_date;
double fixed_asset_cost;
int units_produced_so_far;
char *disposal_date;
char *cost_recovery_period_string;
char *cost_recovery_method;
char *cost_recovery_conversion;
enum depreciation_method depreciation_method =
        depreciation_method_resolve();
int estimated_useful_life_years;
int estimated_useful_life_units;
int estimated_residual_value;
int declining_balance_n;
double finance_accumulated_depreciation;
double tax_adjusted_basis;

/* Process */
FIXED_ASSET_PURCHASE *fixed_asset_purchase_new(
        char *asset_name,
        char *serial_number );

FIXED_ASSET *fixed_asset =
        fixed_asset_fetch(
            asset_name );

char *fixed_asset_purchase_primary_where(
        char *asset_name,
        char *serial_label );

char *fixed_asset_purchase_system_string(
        char *where,
        char *order );

LIST *fixed_asset_purchase_system_list(
        fixed_asset_purchase_system_string(),
        boolean fetch_last_depreciation,
        boolean fetch_last_recovery );

FIXED_ASSET_PURCHASE *fixed_asset_purchase_parse(
        char *input,
        boolean fetch_last_depreciation,
        boolean fetch_last_recovery );

double cost_basis =
        purchase_cost_basis(
            fixed_asset_cost,
            purchase->sales_tax,
            purchase->freight_in,
            fixed_asset_purchase_total(),
            inventory_purchase_total(),
            specific_inventory_purchase_total(),
            supply_expense_purchase_total(),
            prepaid_asset_purchase_total() );

double finance_accumulated_depreciation =
        depreciation_accumulated_depreciation();

double tax_accumulated_depreciation =
        recovery_accumulated_depreciation();

LIST *fixed_asset_purchase_list_depreciate(
        fixed_asset_purchase_list_fetch(
            "finance_accumulated_depreciation <
             cost_basis",
            0 /* not fetch_last_depreciation */,
            0 /* not fetch_last_recovery */ ),
        date_yyyy_mm_dd_now()
            /* depreciation_date */ );

DEPRECIATION *depreciation =
        depreciation_evaluate(
            fixed_asset->asset_name,
            fixed_asset_purchase->serial_label,
            service_placement_date,
            depreciation_prior_depreciation_date(),
            date_yyyy_mm_dd_now()
                /* depreciation_date */,
            fixed_asset_purchase_cost_basis(),
            units_produced_so_far,
            estimated_residual_value,
            estimated_useful_life_years,
            estimated_useful_life_units,
            declining_balance_n,
            finance_accumulated_depreciation
                /* prior_accumulated_depreciation */,
            enum depreciation_method ) );

LIST *fixed_asset_purchase_depreciation_list(
        LIST *fixed_asset_purchase_list );

void fixed_asset_purchase_list_update(
        LIST *fixed_asset_purchase_list );

FILE *fixed_asset_purchase_update_open(
        void );

void fixed_asset_purchase_update(
        update_pipe,
        fixed_asset_purchase_cost_basis(),
        finance_accumulated_depreciation,
        tax_adjusted_basis,
        asset_name,
        serial_label );

void fixed_asset_purchase_finance_fetch_update(
        char *asset_name,
        char *serial_label );

RECOVERY *recovery =
        recovery_evaluate();

void fixed_asset_purchase_cost_fetch_update(
        char *asset_name,
        char *serial_label );

LIST *fixed_asset_purchase_list_cost_recover(
        LIST *fixed_asset_purchase_list,
        int tax_year );

LIST *fixed_asset_purchase_cost_recovery_list(
        LIST *fixed_asset_purchase_list );
```

ENTITY **FIXED_ASSET** **DEPRECIATION** **RECOVERY**

FIXED_ASSET

```
/* Usage */
FIXED_ASSET *fixed_asset_fetch(
      char *asset_name );

/* Attributes */
char *asset_name;
char *account_name;
char *cost_recovery_period_string;
char *cost_recovery_method;
char *cost_recovery_conversion;
double activity_energy_kilowatt_draw;
double activity_depreciation_per_hour;

/* Process */
FIXED_ASSET *fixed_asset_new(
      char *asset_name );

FIXED_ASSET *fixed_asset_parse(
      char *input );

char *fixed_asset_primary_where(
      char *asset_name );

char *fixed_asset_system_string(
      char *where );

char *fixed_asset_name_escape(
      char *fixed_asset );
```

RECOVERY

```
/* Usage */
RECOVERY *recovery_evaluate(
    char *asset_name,
    char *serial_label,
    int tax_year,
    double fixed_asset_purchase->
        cost_basis,
    char *fixed_asset_purchase->
        service_placement_date,
    char *fixed_asset_purchase->
        disposal_date,
    char *fixed_asset_purchase->
        cost_recovery_period_string,
    char *fixed_asset_purchase->
        cost_recovery_method,
    char *fixed_asset_purchase->
        cost_recovery_conversion );

RECOVERY *recovery_fetch(
    char *asset_name,
    char *serial_label,
    int tax_year );

LIST *recovery_list_fetch(
    char *asset_name,
    char *serial_label );

void recovery_fixed_asset_list_set(
    LIST *fixed_asset_list,
    int tax_year );

/* Input */
char *asset_name;
char *serial_label;
int tax_year;
double cost_basis;
char *service_placement_date;
char *disposal_date;
char *cost_recovery_period_string;
char *cost_recovery_method;
char *cost_recovery_conversion;

/* Process */
char *recovery_system_string(
    char *where );

LIST *recovery_system_list(
    recovery_system_string() );

char *recovery_primary_where(
    char *asset_name,
    char *serial_label,
    int tax_year );

double recovery_period_years(
    char *cost_recovery_period_string );

int service_placement_month =
    date_month(
        date_yyyy_mm_dd_new(
            service_placement_date ) );

int service_placement_year =
    date_year(
        date_yyyy_mm_dd_new(
            service_placement_date ) );

int disposal_month =
    date_month(
        date_yyyy_mm_dd_new(
            disposal_date ) );

int disposal_year =
    date_year(
        date_yyyy_mm_dd_new(
            disposal_date ) );

int recovery_service_placement_years(
    int tax_year,
    int service_placement_year )
{
    return A - B + 1;
}

char *recovery_subquery_where(
    int tax_year );

RECOVERY_STRAIGHT_LINE *straight_line =
    recovery_statutory_straight_line_evaluate(
        tax_year,
        cost_basis,
        service_placement_month,
        service_placement_year,
        disposal_month,
        disposal_year,
        recovery_period_years(),
        cost_recovery_conversion );

RECOVERY_ACCELERATED *accelerated =
    recovery_statutory_accelerated_evaluate(
        tax_year,
        cost_basis,
        service_placement_month,
        service_placement_year,
        disposal_month,
        disposal_year,
        recovery_period_years(),
        cost_recovery_conversion );

void recovery_fixed_asset_list_display(
    FILE *output_pipe,
    LIST *fixed_asset_list );

void recovery_list_insert(
    LIST *recovery_list );

double recovery_rate(
    straight_line,
    accelerated );

double recovery_amount(
    straight_line,
    accelerated );
```

RECOVERY_STRAIGHT_LINE **RECOVERY_ACCELERATED**

RECOVERY_STRAIGHT_LINE

```
/* Usage */
RECOVERY_STRAIGHT_LINE *
    recovery_statutory_straight_line_evaluate(
        int tax_year,
        double cost_basis,
        int service_placement_month,
        int service_placement_year,
        int disposal_month,
        int disposal_year,
        double recovery_period_years,
        char *cost_recovery_conversion );

/* Process */
RECOVERY_STRAIGHT_LINE *recovery_straight_line_calloc();

double service_placement_years =
    recovery_service_placement_years(
        tax_year,
        service_placement_year );

double straight_line_rate =
    recovery_statutory_straight_line_half_year_rate(
        int recovery_service_placement_years(),
        int tax_year,
        int disposal_year,
        double recovery_period_years );

double recovery_statutory_straight_line_half_year_mid_year_rate(
    double recovery_period_years );

double recovery_statutory_straight_line_half_year_book_ends_rate(
    double recovery_statutory_straight_line_mid_year_rate() );

double straight_line_amount =
    recovery_straight_line_amount(
        double straight_line_rate,
        double cost_basis )
{
    return A X B;
}
```

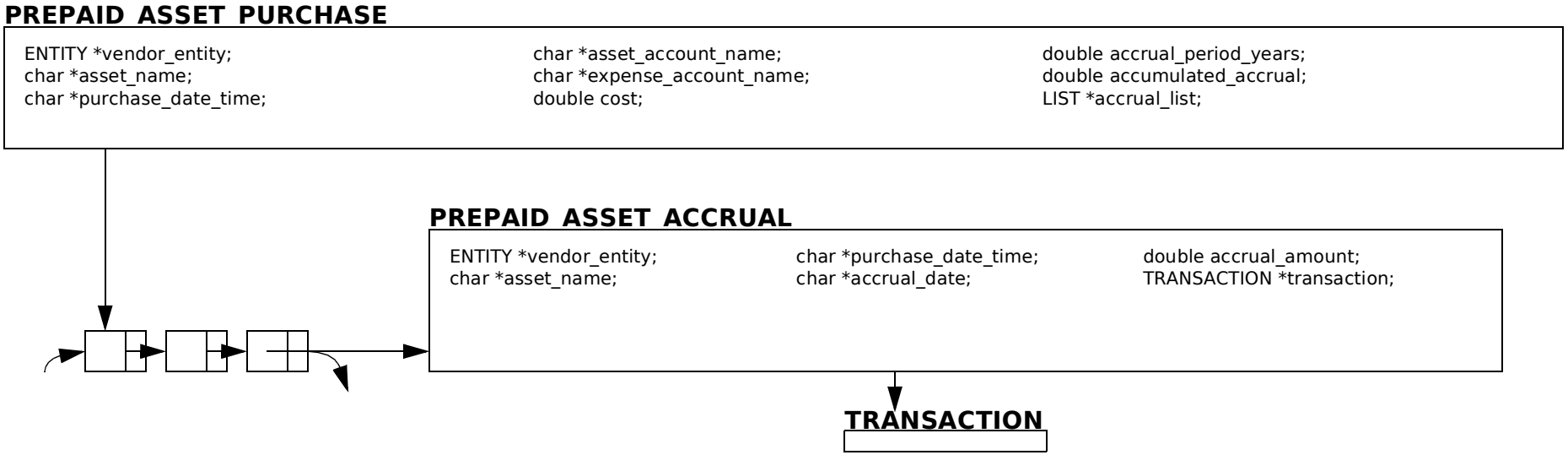
RECOVERY_ACCELERATED

```
/* Usage */
RECOVERY_ACCELERATED *
    recovery_statutory_accelerated_evaluate(
        int tax_year,
        double cost_basis,
        int service_placement_month,
        int service_placement_year,
        int disposal_month,
        int disposal_year,
        double recovery_period_years,
        char *cost_recovery_conversion );

/* Process */
RECOVERY_ACCELERATED *recovery_accelerated_calloc();

double accelerated_rate =
    recovery_statutory_accelerated_half_year_rate(
        int recovery_service_placement_years(),
        int tax_year,
        int disposal_year,
        double recovery_period_years );

double recovery_statutory_accelerated_half_year_evaluate(
    int service_placement_years,
    double recovery_period_years );

double accelerated_amount =
    recovery_accelerated_amount(
        double accelerated_rate,
        double cost_basis );
```

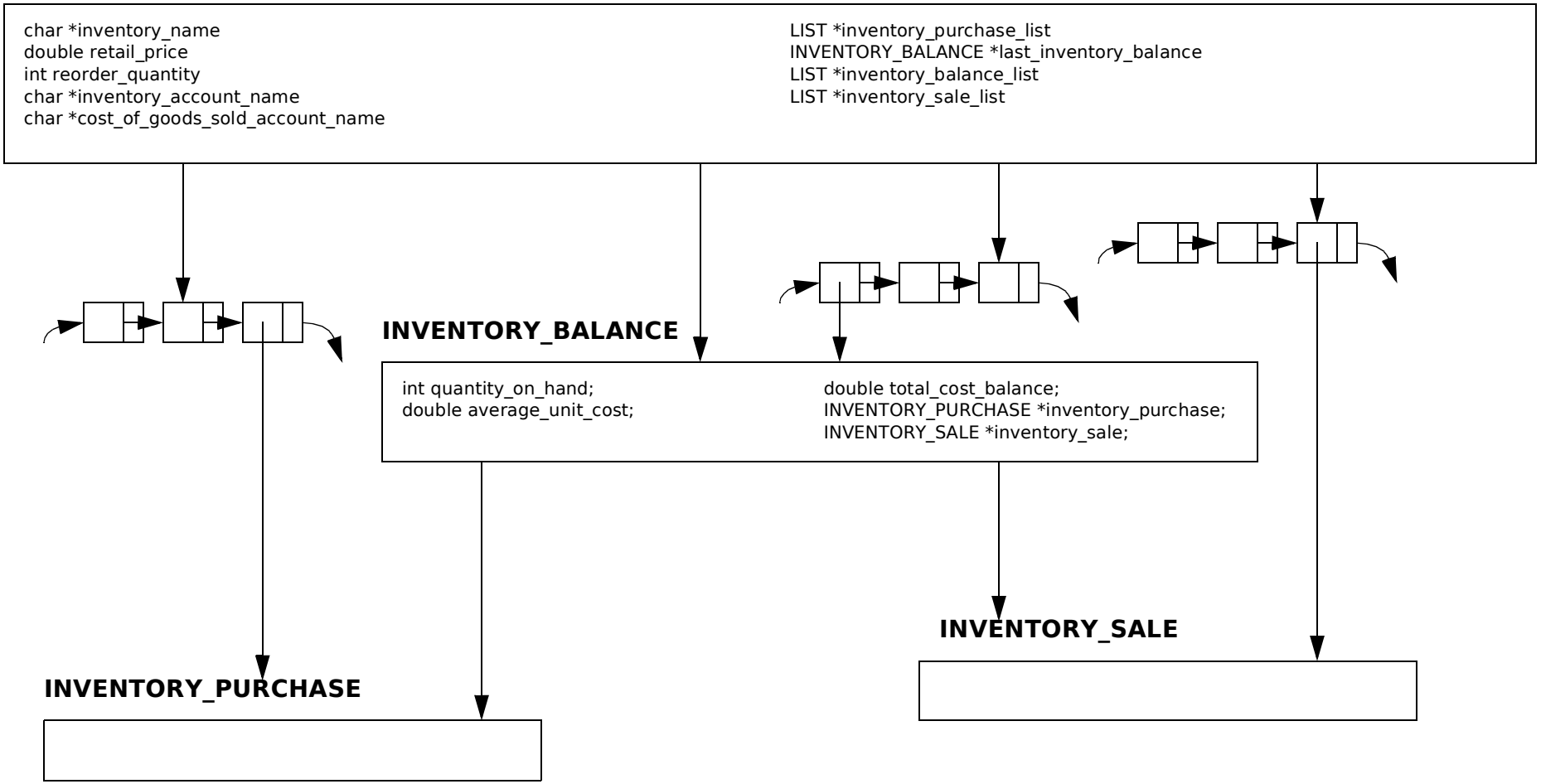
DEPRECIATION

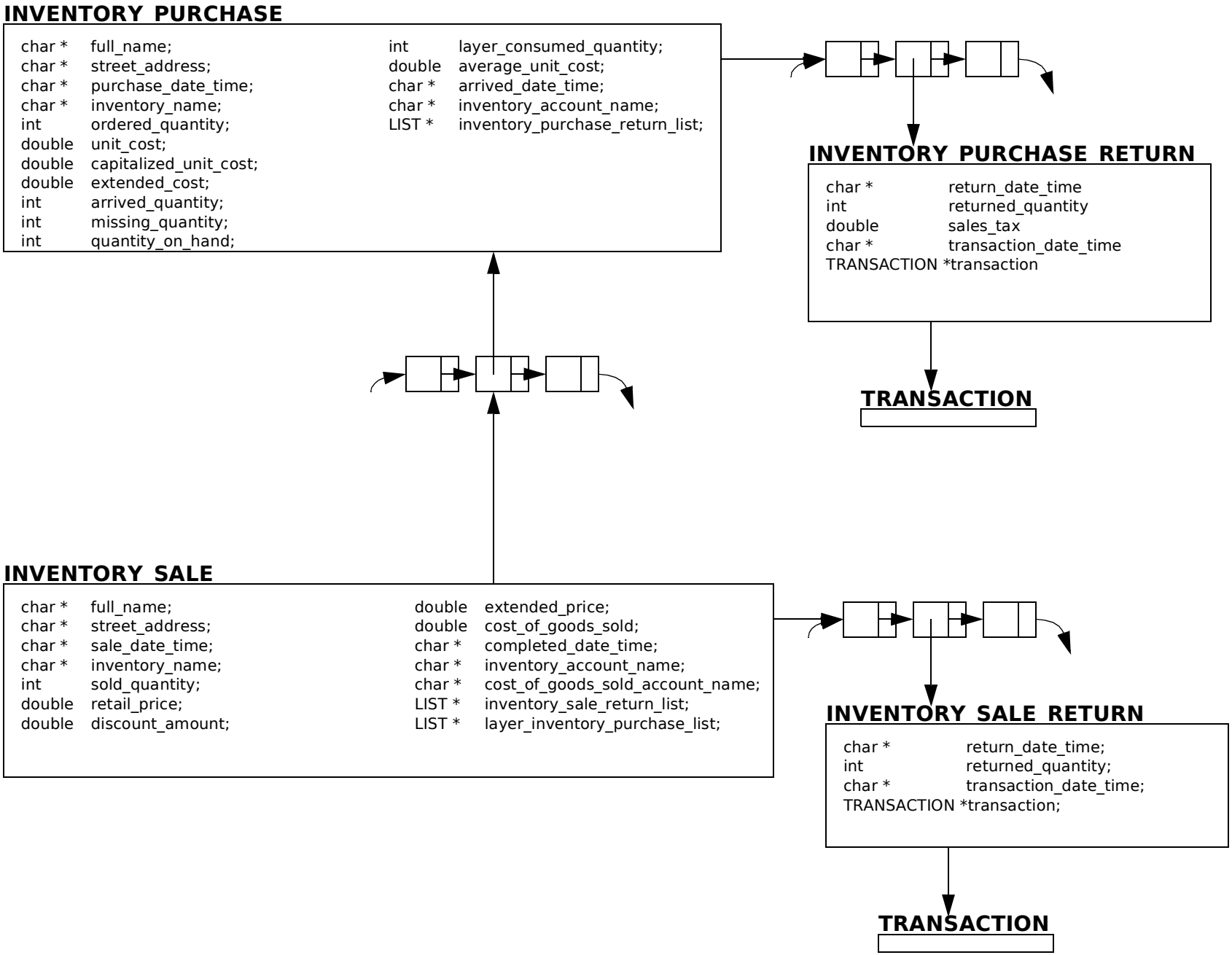
```
/* Usage */
DEPRECIATION *depreciation_evaluate(
        char *asset_name,
        char *serial_label,
        depreciation_method_evaluate(),
        char *service_placement_date,
        char *depreciation_prior_depreciation_date(),
        char *date_yyyy_mm_dd_now()
                /* depreciation_date */,
        double cost_basis,
        int units_produced_so_far,
        int estimated_residual_value,
        int estimated_useful_life_years,
        int estimated_useful_life_units,
        int declining_balance_n,
        double fixed_asset_purchase->
                finance_accumulated_depreciation
                  /* prior_accumulated_depreciation */ );

DEPRECIATION *depreciation_fetch(
        char *asset_name,
        char *serial_label,
        char *depreciation_date );

/* Process */
enum depreciation_method =
        depreciation_method_evaluate(
                depreciation_method_string );

char *depreciation_prior_depreciation_date(
        char *asset_name,
        char *serial_label,
        char *current_depreciation_date );

double depreciation_amount(
        enum depreciation_method,
        char *service_placement_date,
        depreciation_prior_depreciation_date(),
        char *depreciation_date,
        double cost_basis,
        depreciation_units_produced_current(),
        int estimated_residual_value,
        int estimated_useful_life_years,
        int estimated_useful_life_units,
        int declining_balance_n,
        double prior_accumulated_depreciation );

double depreciation_straight_line(
        char *service_placement_date,
        char *prior_depreciation_date,
        char *depreciation_date,
        double cost_basis,
        double estimated_residual_value,
        int estimated_useful_life_years,
        double prior_accumulated_depreciation );

double depreciation_sum_of_years_digits(
        char *service_placement_date,
        char *prior_depreciation_date,
        char *depreciation_date,
        double cost_basis,
        double estimated_residual_value,
        int estimated_useful_life_years,
        prior_accumulated_depreciation );

double depreciation_double_declining_balance(
        char *service_placement_date,
        char *prior_depreciation_date,
        char *depreciation_date,
        double cost_basis,
        double estimated_residual_value,
        int estimated_useful_life_years,
        double prior_accumulated_depreciation );

double depreciation_n_declining_balance(
        char *service_placement_date,
        char *prior_depreciation_date,
        char *depreciation_date,
        double cost_basis,
        double estimated_residual_value,
        int estimated_useful_life_years,
        int declining_balance_n,
        double prior_accumulated_depreciation );

int depreciation_units_produced_total(
        char *asset_name,
        char *serial_label );

int depreciation_units_produced_current(
        int units_produced_so_far,
        depreciation_units_produced_total() );

double depreciation_units_of_production(
        double cost_basis,
        double estimated_residual_value,
        int estimated_useful_life_units,
        int depreciation_units_produced_current(),
        double prior_accumulated_depreciation );

double depreciation_fraction_of_year(
        char *service_placement_date,
        char *prior_depreciation_date,
        char *depreciation_date_string );

double depreciation_accumulated_depreciation(
        prior_accumulated_depreciation,
        depreciation_amount() );

char *depreciation_subquery_where(
        char *depreciation_date );

TRANSACTION *depreciation_transaction(
        entity_self->entity->full_name,
        entity_self->entity->street_address,
        predictive_transaction_date_time(
                depreciation_date ),
        depreciation_amount(),
        account_depreciation_expense(),
        account_accumulated_depreciation() );

void depreciation_insert(
        FILE *insert_pipe,
        char *asset_name,
        char *serial_label,
        char *depreciation_date,
        int depreciation_units_produced_current(),
        double depreciation_amount,
        char *depreciation_transaction()->
                transaction_date_time );

void depreciation_list_negate_depreciation_amount(
        LIST *depreciation_list );

LIST *depreciation_transaction_list(
        fixed_asset_purchase_depreciation_list() );
```

straight_line,
double_declining_balance,
n_declining_balance,
sum_of_years_digits,
units_of_production,
not_depreciated


PREPAID_ASSET_PURCHASE
ENTITY *vendor_entity;
char *asset_name;
char *purchase_date_time;
char *asset_account_name;
char *expense_account_name;
double cost;
double accrual_period_years;
double accumulated_accrual;
LIST *accrual_list;
PREPAID_ASSET_ACCRUAL
ENTITY *vendor_entity;
char *asset_name;
char *purchase_date_time;
char *accrual_date;
double accrual_amount;
TRANSACTION *transaction;
TRANSACTION

INVENTORY


char *inventory_name
double retail_price
int reorder_quantity
char *inventory_account_name
char *cost_of_goods_sold_account_name
LIST *inventory_purchase_list
INVENTORY_BALANCE *last_inventory_balance
LIST *inventory_balance_list
LIST *inventory_sale_list
INVENTORY_BALANCE
int quantity_on_hand;
double average_unit_cost;
double total_cost_balance;
INVENTORY_PURCHASE *inventory_purchase;
INVENTORY_SALE *inventory_sale;
INVENTORY_SALE
INVENTORY_PURCHASE

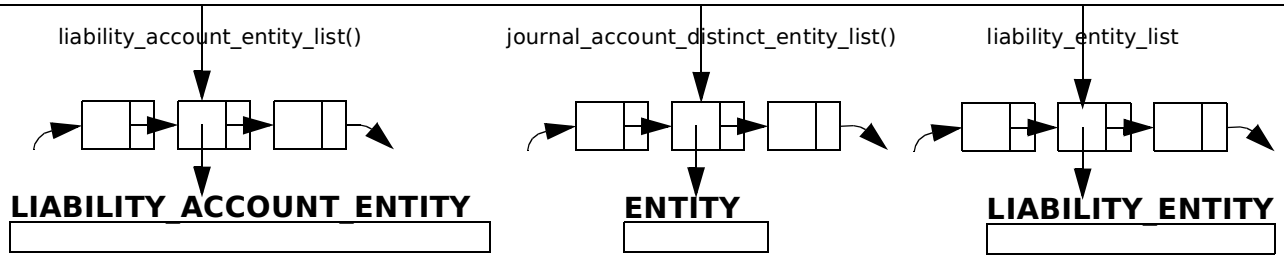

INVENTORY_PURCHASE
char * full_name;
char * street_address;
char * purchase_date_time;
char * inventory_name;
int ordered_quantity;
double unit_cost;
double capitalized_unit_cost;
double extended_cost;
int arrived_quantity;
int missing_quantity;
int quantity_on_hand;
int layer_consumed_quantity;
double average_unit_cost;
char * arrived_date_time;
char * inventory_account_name;
LIST * inventory_purchase_return_list;
INVENTORY_PURCHASE_RETURN
char * return_date_time
int returned_quantity
double sales_tax
char * transaction_date_time
TRANSACTION *transaction
TRANSACTION
INVENTORY_SALE
char * full_name;
char * street_address;
char * sale_date_time;
char * inventory_name;
int sold_quantity;
double retail_price;
double discount_amount;
double extended_price;
double cost_of_goods_sold;
char * completed_date_time;
char * inventory_account_name;
char * cost_of_goods_sold_account_name;
LIST * inventory_sale_return_list;
LIST * layer_inventory_purchase_list;
INVENTORY_SALE_RETURN
char * return_date_time;
int returned_quantity;
char * transaction_date_time;
TRANSACTION *transaction;
TRANSACTION

LIABILITY_CALCULATE

```
/* Usage */
LIABILITY_CALCULATE *liability_calculate_new(
   char *application_name );

/* Process */
LIABILITY_CALCULATE *liability_calculate_calloc( void );

LIST *liability_account_entity_list();

LIST *account_current_liability_name_list(
   ACCOUNT_TABLE,
   SUBCLASSIFICATION_CURRENT_LIABILITY,
   ACCOUNT_UNCLEARED_CHECKS );

LIST *journal_account_distinct_entity_list(
   JOURNAL_TABLE,
   account_current_liability_name_list()
         /* account_name_list */ );
```

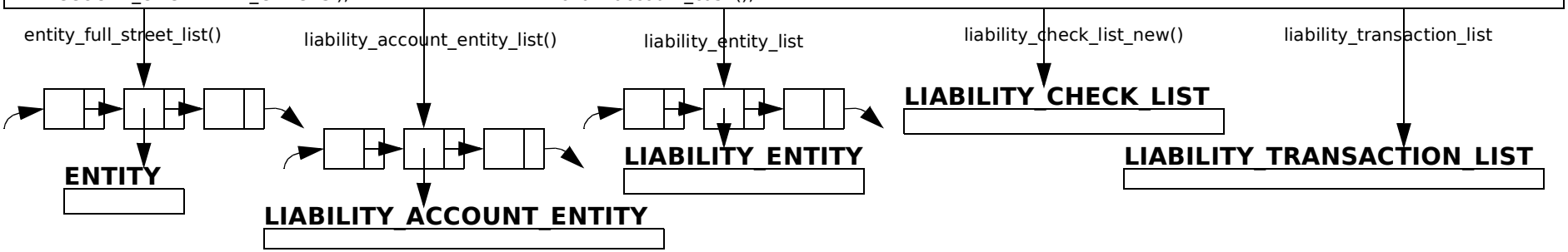
```
if ( !list_rewind( journal_account_distinct_entity_list() )
{
   return (LIABILITY_CALCULATE *)0;
}

LIST *account_receivable_name_list(
   ACCOUNT_TABLE,
   SUBCLASSIFICATION_RECEIVABLE );

LIST *liability_entity_list = list_new();

for ENTITY *entity in journal_account_distinct_entity_list
{
   list_set(
         liability_entity_list,
         liability_entity_new(
               liability_account_entity_list(),
               account_current_liability_name_list(),
               account_receivable_name_list(),
```

```
               entity ) );
}

if ( !list_length( liability_entity_list ) )
{
   return (LIABILITY_CALCULATE *)0;
}

/* Driver */
void liability_calculate_stdout(
   LIST *liability_entity_list );

/* Process */
for LIABILITY_ENTITY *liability_entity in liability_entity_list
{
   printf(    “%s\n”,
               liability_entity_display(
                     liability_entity );
}
```



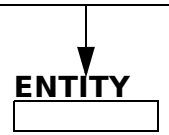
LIABILITY_PAYMENT

```
/* Usage */
LIABILITY_PAYMENT *liability_payment_new(
   char *application_name,
   double dialog_box_payment_amount,
   int starting_check_number,
   char *memo,
   char *document_root_directory,
   char *process_name,
   char *session_key,
   LIST *entity_full_street_list(
         full_name_list,
         street_address_list ) );

/* Process */
LIABILITY_PAYMENT *liability_payment_calloc( void );

if ( !list_length( entity_full_street_list() )
{
   return (LIABILITY_PAYMENT *)0;
}

char *transaction_memo( memo );

/* For dialog_box_payment_amount or memo
only one entity at a time. */

if ( list_length( entity_full_street_list() > 1
&& (    dialogy_box_payment_amount
||      *transaction_memo() ) )
{
   return (LIABILITY_PAYMENT *)0;
}

LIST *liability_account_entity_list();

LIST *account_current_liability_name_list(
   ACCOUNT_TABLE,
   SUBCLASSIFICATION_CURRENT_LIABILITY,
   ACCOUNT_UNCLEARED_CHECKS );

LIST *account_receivable_name_list(
   ACCOUNT_TABLE,
   SUBCLASSIFICATION_RECEIVABLE );

LIST *liability_entity_list = list_new();

for ENTITY *entity in entity_full_street_list
{
   list_set(
         liability_entity_list,
         liability_entity_new(
               liability_account_entity_list(),
               account_current_liability_name_list(),
               account_receivable_name_list(),
               entity ) );
}

if ( !list_length( liability_entity_list ) )
{
   exit( 1 );
}

if ( starting_check_number )
{
   LIABILITY_CHECK_LIST *liability_check_list =
         liability_check_list_new(
               application_name,
               starting_check_number,
               transaction_memo(),
               document_root_directory,
               process_name,
               session_key,
               liability_entity_list );
}

char *liability_payment_credit_account_name(
   int starting_check_number,
   char *account_cash(),
   char *account_uncleared_checks() );

LIABILITY_TRANSACTION_LIST *liability_transaction_list =
   liability_transaction_list_new(
         dialog_box_payment_amount,
         starting_check_number,
         transaction_memo(),
         liability_entity_list,
         liability_payment_credit_account_name() );

if ( !liability_transaction_list )
{
   exit( 1 );
}

/* Driver */
if ( liability_check_list )
{
      appaserver_library_output_ftp_prompt(
            liability_check_list->
                  liability_check_appaserver_link->
                  ftp_filename,
            LIABILITY_PROMPT,
            "check_page" /* target_frame */,
            (char *)0 /* application_type */ );
}

if ( execute )
{
      /* May reset transaction_date_time */
      void liability_transaction_list_insert(
            liability_transaction_list,
            appaserver_error_get_filename(
                  application_name ) );
}

void liability_transaction_list_html_display(
      liability_transaction_list );
```



LIABILITY_ACCOUNT_ENTITY

```
/* Usage */
LIST *liability_account_entity_list( void );

/* Process */
char *liability_account_entity_system_string(
      char *LIABILITY_ACCOUNT_ENTITY_SELECT,
      char *LIABILITY_ACCOUNT_ENTITY_TABLE );

LIST *liability_account_entity_system_list(
      char *liability_account_entity_system_string() );

LIABILITY_ACCOUNT_ENTITY *
      liability_account_entity_parse(
            char *string_input() );

LIABILITY_ACCOUNT_ENTITY *
      liability_account_entity_new(
            char *strdup( account_name ) );

LIABILITY_ACCOUNT_ENTITY *
      liability_account_entity_calloc(
            void );

ENTITY *entity =
      entity_new(
            strdup( full_name ),
            strdup( street_address ) );

/* Attributes */
char *account_name;
ENTITY *entity;

/* Public */
LIABILITY_ACCOUNT_ENTITY *
      liability_account_entity_seek_account(
            char *account_name,
            LIST *liability_account_entity_list() );

LIABILITY_ACCOUNT_ENTITY *
      liability_account_entity_seek_entity(
            char *full_name,
            char *street_address,
            LIST *liability_account_entity_list() );
```



LIABILITY_CHECK_LIST

```
/* Usage */
LIABILITY_CHECK_LIST *liability_check_list_new(
        char *application_name,
        int starting_check_number,
        char *transaction_memo(),
        char *document_root_directory,
        char *process_name,
        char *session_key,
        LIST *liability_entity_list() );

/* Process */
LIABILITY_CHECK_LIST *liability_check_list_calloc( void );

char *liability_check_list_documentclass( void );

char *liability_check_list_usepackage( void );

char *liability_check_list_pagenumbering_gobble( void );

char *liability_check_list_begin_document( void );

char *liability_check_list_heading(
        char *liability_check_list_documentclass(),
        char *liability_check_list_usepackage(),
        char *liability_check_list_pagenumbering_gobble(),
        char *liability_check_list_begin_document() );
```

```
LIST *list = list_new();

for LIABILITY_ENTITY *liability_entity in
        liability_enitity_list
{
        list_set(
                list,
                liability_check_new(
                        starting_check_number++
                                /* check_number */,
                        transaction_memo,
                        liability_entity ) );
}

char *liability_check_list_end_document( void );

LIABILITY_CHECK_APPASERVER_LINK *
        liability_check_appaserver_link =
                liability_check_appaserver_link_new(
                        application_name,
                        document_root_directory,
                        process_name,
                        session_key );

FILE *output_file =
        fopen(
                liability_check_appaserver_link->
                        output_filename,
                "w" );

fprintf(    output_file,
            "%s\n",
            liability_check_list_heading() );

for LIABILITY_CHECK *liability_check in list
{
        fprintf(    output_file,
                    "%s\n",
                    liability_check->output_string );
}

fprintf(    output_file,
            "%s\n",
            liability_check_list_end_document() );

fclose( output_file );

latex_tex2pdf(
        liability_check_appaserver_link->
                output_filename,
        appaserver_link_source_directory(
                document_root_directory,
                application_name )
                        /* working_directory */ );
```

LIABILITY_CHECK_APPASERVER_LINK

```
/* Usage */
LIABILITY_CHECK_APPASERVER_LINK *
    liability_check_appaserver_link_new(
    char *application_name,
    char *document_root_directory,
    char *process_name,
    char *session_key );

/* Process */
LIABILITY_CHECK_APPASERVER_LINK *
    liability_check_appaserver_link_calloc(
        void );

APPASERVER_LINK_FILE *appaserver_link_file =
    appaserver_link_file_new(
        application_http_prefix( application_name ),
        appaserver_library_get_server_address(),
        ( application_prepend_http_protocol_yn(
            application_name ) == 'y' ),
        document_root_directory,
        process_name /* filename_stem */,
        application_name,
        getpid(),
        (char *)0 /* session_key */,
         "tex" );

char *output_filename =
    appaserver_link_get_output_filename(
        appaserver_link_file->
            output_file->
            document_root_directory,
        appaserver_link_file->application_name,
        appaserver_link_file->filename_stem,
        appaserver_link_file->begin_date_string,
        appaserver_link_file->end_date_string,
        appaserver_link_file->process_id,
        appaserver_link_file->session,
        appaserver_link_file->extension );

appaserver_link_file->extension = "pdf";

char *pdf_output_filename =
    appaserver_link_get_link_prompt(
        appaserver_link_file->
            link_prompt->
            prepend_http_boolean,
        appaserver_link_file->
            link_prompt->
            http_prefix,
        appaserver_link_file->
            link_prompt->
            server_address,
        appaserver_link_file->application_name,
        appaserver_link_file->filename_stem,
        appaserver_link_file->begin_date_string,
        appaserver_link_file->end_date_string,
        appaserver_link_file->process_id,
        appaserver_link_file->session,
        appaserver_link_file->extension );

char *ftp_filename =
    appaserver_link_get_link_prompt(
        appaserver_link_file->
            link_prompt->
            prepend_http_boolean,
        appaserver_link_file->
            link_prompt->
            http_prefix,
        appaserver_link_file->
            link_prompt->
            server_address,
        appaserver_link_file->application_name,
        appaserver_link_file->filename_stem,
        appaserver_link_file->begin_date_string,
        appaserver_link_file->end_date_string,
        appaserver_link_file->process_id,
        appaserver_link_file->session,
        appaserver_link_file->extension );

appaserver_link_file->extension = "pdf";
appaserver_link_file->session = session_key;
appaserver_link_file->process_id = 0;

char *document_root_filename =
    appaserver_link_get_output_filename(
        appaserver_link_file->
            output_file->
            document_root_directory,
        appaserver_link_file->application_name,
        appaserver_link_file->filename_stem,
        appaserver_link_file->begin_date_string,
        appaserver_link_file->end_date_string,
        appaserver_link_file->process_id,
        appaserver_link_file->session,
        appaserver_link_file->extension );
```

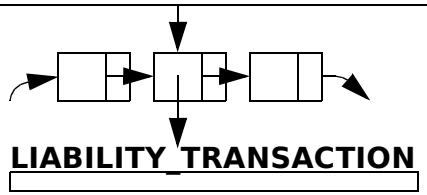
APPASERVER_LINK_FILE

LIABILITY_CHECK

```
/* Usage */
LIABILITY_CHECK *liability_check_new(
	int check_number,
	char *transaction_memo(),
	LIABILITY_ENTITY *liability_entity );

/* Process */
LIABILITY_CHECK *liability_check_calloc( void );

char *liability_check_dollar_text(
	double liability_entity->amount_due );

char *liability_check_escape_payable_to(
	char *liability_entity->entity->full_name );

char *liability_check_move_down( void );

char *liability_check_date_display(
	char *pipe2string(
		char *LIABILITY_CHECK_DATE_COMMAND )
			/* check_date */ );

char *liability_check_amount_due_display(
	char *place_commas_in_money(
		double liability_entity->amount_due )
			/* amount_due_display */ );

char *liability_check_vendor_name_amount_due_display(
	char *liability_check_escape_payable_to(),
	char *liability_check_amount_due_display() );

char *liability_check_amount_due_stub_display(
	char *liability_check_amount_due_display() );

char *liability_check_dollar_text_display(
	char *liability_check_dollar_text() );

char *liability_check_memo_display(
	char *transaction_memo );

char *liability_check_number_display(
	int check_number );

char *liability_check_newpage( void );

char *liability_check_output_string(
	char *liability_check_move_down(),
	char *liability_check_date_display(),
	char *liability_check_vendor_name_amount_due_display(),
	char *liability_check_amount_due_stub_display(),
	char *liability_check_dollar_text_display(),
	char *liability_check_memo_display(),
	char *liability_check_number_display(),
	char *liability_check_newpage() );
```

LIABILITY_TRANSACTION_LIST

```
/* Usage */
LIABILITY_TRANSACTION_LIST *
   liability_transaction_list_new(
         double dialog_box_payment_amount,
         int starting_check_number,
         char *transaction_memo(),
         LIST *liability_entity_list,
         char *liability_payment_credit_account_name );

/* Process */
if ( dialog_box_payment_amount
&& list_length( liability_entity_list ) > 1 )
{
   return (LIABILITY_TRANSACTION_LIST *)0;
}

LIABILITY_TRANSACTION_LIST *
   liability_transaction_list_calloc(
         void );

LIST *list = list_new();

DATE *transaction_date_time =
   date_now_new(
         date_utc_offset() );

for LIABILITY_ENTITY *liability_entity in liability_entity_list
{
   list_set(
         list,
         liability_transaction_new(
               dialog_box_payment_amount,
               starting_check_number
                     /* check_number */,
               transaction_memo,
               liability_payment_credit_account_name,
               liability_entity,
               transaction_date_time ) );

   if ( starting_check_number )
   {
         starting_check_number++;
   }

   date_increment_seconds( transaction_date_time, 1 );
}

/* Usage */
/* May reset transaction_date_time */
void liability_transaction_list_insert(
   LIABILITY_TRANSACTION_LIST *
         liability_transaction_list,
   char *appaserver_error_filename );

/* Process */
LIST *liability_transaction_list_extract(
   liability_transaction_list->list );

TRANSACTION_LIST *transaction_list =
   transaction_list_new(
         LIST *liability_transaction_list_extract() );

void transaction_list_insert(
   transaction_list,
   appaserver_error_filename );

/* Usage */
void liability_transaction_list_html_display(
   LIABILITY_TRANSACTION_LIST *
         liability_transaction_list );

/* Process */
LIST *liability_transaction_list_extract(
   liability_transaction_list->list );

TRANSACTION_LIST *transaction_list =
   transaction_list_new(
         LIST *liability_transaction_list_extract() );

void transaction_list_html_display(
   transaction_list );

/* Private */
LIST *liability_transaction_list_extract(
   LIST *liability_transaction_list->list );
```



LIABILITY_TRANSACTION

```
/* Usage */
LIABILITY_TRANSACTION *liability_transaction_new(
        double dialog_box_payment_amount,
        int check_number,
        char *transaction_memo(),
        char *liability_payment_credit_account_name,
        LIABILITY_ENTITY *liability_entity,
        DATE *transaction_date_time );

/* Process */
LIABILITY_TRANSACTION *liability_transaction_calloc(
        void );
```

```
LIABILITY_JOURNAL_LIST *liability_journal_list =
        liability_journal_list_new(
                dialog_box_payment_amount,
                liability_payment_credit_account_name,
                liability_entity );

if ( !liability_journal_list )
{
        return (LIABILITY_TRANSACTION *)0;
}

double journal_list_transaction_amount(
        liability_journal_list->journal_list );
```

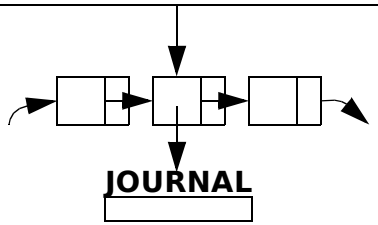
```
TRANSACTION *transaction =
        transaction_entity_new(
                liability_entity->entity,
                date_display_19( transaction_date_time ),
                journal_list_transaction_amount(),
                check_number,
                (*transaction_memo)
                        ? transaction_memo
                        : LIABILITY_TRANSACTION_MEMO,
                liability_journal_list->journal_list );
```

LIABILITY_JOURNAL_LIST

TRANSACTION

LIABILITY_JOURNAL_LIST

```
/* Usage */
LIABILITY_JOURNAL_LIST *liability_journal_list_new(
	double dialog_box_payment_amount,
	char *liability_payment_credit_account_name,
	LIABILITY_ENTITY *liability_entity );

/* Process */
LIABILITY_JOURNAL_LIST *liability_journal_list_calloc(
	void );

LIST *journal_list = list_new();

if ( dialog_box_payment_amount )
{
	liability_journal_list_transaction_amount(
		dialog_box_payment_amount );
}
else
{
	liability_journal_list_transaction_amount(
		liability_entity->amount_due );
}

list_set(
	journal_list,
	journal_credit_new(
		liability_payment_credit_account_name,
		liability_journal_list_transaction_amount() ) );

for LIABILITY_ACCOUNT *liability_account in
	liability_entity->liability->liability_account_list->list
{
	list_set(
		journal_list,
		journal_debit_new(
			liability_account->account_name,
			(dialog_box_payment_amount)
				? dialog_box_payment_amount
				: liability_account->credit_amount ) );
}

if ( liability_entity->receivable )
{
	for RECEIVABLE_ACCOUNT *receivable_account in
		liability_entity->receivable->receivable_account_list->list
	{
		list_set(
			journal_list,
			journal_credit_new(
				receivable_account->account_name,
				receivable_account->debit_amount ) );
	}
}

/* Private */
double liability_journal_list_transaction_amount(
	double amount_due );
```



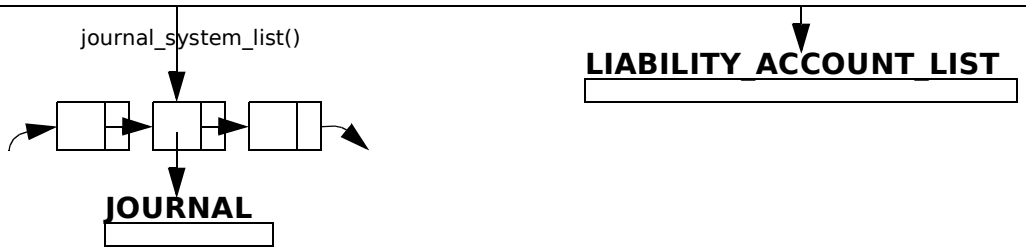
LIABILITY_ENTITY

```
/* Usage */
LIABILITY_ENTITY *liability_entity_new(
   LIST *liability_account_entity_list(),
   LIST *account_current_liability_name_list(),
   LIST *account_receivable_name_list(),
   ENTITY *entity );

/* Process */
if ( !entity ) exit( 1 );

LIABILITY_ENTITY *liability_entity_calloc( void );

LIABILITY_ACCOUNT_ENTITY *
      liability_account_entity =
            liability_account_entity_seek_entity(
                  entity->full_name,
                  entity->street_address,
                  liability_account_entity_list );

if ( liability_account_entity )
{
      LIABILITY *liability =
            liability_account_fetch(
                  liability_account_entity->
                        accout_name );
}
else
{
      LIABILITY *liability =
            liability_entity_fetch(
                  entity->full_name,
                  entity->street_address,
                  account_current_liability_name_list );
}

if ( !liability )
{
   return (LIABILITY_ENTITY *)0;
}

if ( !liability_account_entity )
{
      RECEIVABLE *receivable =
            receivable_fetch(
                  entity->full_name,
                  entity->street_address,
                  account_receivable_name_list );
}

double liability_entity_amount_due(
      LIABILITY *liability,
      RECEIVABLE *receivable );

/* Driver */
char *liability_entity_display(
      LIABILITY_ENTITY *liability_entity );
```

LIABILITY_ACCOUNT_ENTITY **LIABILITY** **RECEIVABLE**

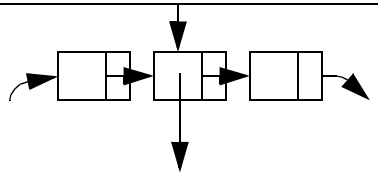
LIABILITY

```
/* Usage */
LIABILITY *liability_entity_fetch(
        char *full_name,
        char *street_address,
        LIST *account_current_liability_name_list() );

/* Process */
LIABILITY *liability_calloc();

char *timlib_in_clause(
        account_current_liability_name_list );

char *liability_entity_where(
        char *full_name,
        char *street_address,
        char *timlib_in_clause() );

LIST *journal_system_list(
        journal_system_string(
                JOURNAL_SELECT,
                JOURNAL_TABLE,
                liability_entity_where() ),
        0 /* not fetch_check_number */,
        0 /* not fetch_memo */ );

if ( !list_length( journal_system_list() )
{
        return (LIABILITY *)0;
}

double journal_credit_debit_difference_sum(
        journal_system_list() );

if ( !journal_credit_debit_difference_sum() )
{
        return (LIABILITY *)0;
}

LIABILITY_ACCOUNT_LIST *
        liability_account_list =
                liability_account_list_new(
                        journal_system_list() );

/* Usage */
LIABILITY *liability_account_fetch(
        char *liability_accout_name );

/* Process */
LIABILITY *liability_calloc();

char *liability_account_where(
        char *liability_account_name );

LIST *journal_system_list(
        journal_system_string(
                JOURNAL_SELECT,
                JOURNAL_TABLE,
                liability_account_where() ),
        0 /* not fetch_check_number */,
        0 /* not fetch_memo */ );

if ( !list_length( journal_system_list() )
{
        return (LIABILITY *)0;
}

double journal_credit_debit_difference_sum(
        journal_system_list() );

if ( !journal_credit_debit_difference_sum() )
{
        return (LIABILITY *)0;
}

LIABILITY_ACCOUNT_LIST *
        liability_account_list =
                liability_account_list_new(
                        journal_system_list() );

/* Private */
LIABILITY *liability_calloc( void );
```



LIABILITY_ACCOUNT_LIST

```
/* Usage */
LIABILITY_ACCOUNT_LIST *
    liability_account_list_new(
        LIST *journal_system_list() );

/* Process */
LIABILITY_ACCOUNT_LIST *
    liability_account_list_calloc(
        void );

LIST *list = list_new();

for JOURNAL *journal in journal_system_list
{
    LIABILITY_ACCOUNT *
        liability_account =
            liability_account_getset(
                list,
                journal->account_name );

    if ( journal->credit_amount )
    {
        liability_account->credit_amount +=
            journal->credit_amount;
    }
    else
    {
        liability_account->credit_amount -=
            journal->debit_amount;
    }
}
```



LIABILITY_ACCOUNT

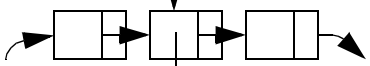
```
/* Usage */
LIABILITY_ACCOUNT *
    liability_account_getset(
        LIST *list,
        char *account_name );

/* Process */
for LIABILITY_ACCOUNT *liability_account in list
{
    if ( strcmp(
        account_name,
        liability_account->account_name ) == 0 )
    {
        return liability_account;
    }
}

LIABILITY_ACCOUNT *
    liability_account =
        liability_account_new(
            account_name );

LIABILITY_ACCOUNT *
    liability_account_calloc(
        void );

list_set(
    list,
    liability_account );

/* Attributes */
char *account_name;
double credit_amount;
```

RECEIVABLE

```
/* Usage */
RECEIVABLE *receivable_fetch(
        char *full_name,
        char *street_address,
        LIST *account_receivable_name_list() );

/* Process */
RECEIVABLE *receivable_calloc( void );

char *timlib_in_clause(
        account_receivable_name_list );

char *receivable_where(
        char *full_name,
        char *street_address,
        char *timlib_in_clause() );

LIST *journal_system_list(
        journal_system_string(
                JOURNAL_SELECT,
                JOURNAL_TABLE,
                receivable_where() ),
        0 /* not fetch_check_number */,
        0 /* not fetch_memo */ );

if ( !list_length( journal_system_list() )
{
        return (RECEIVABLE *)0;
}

double journal_debit_credit_difference_sum(
        journal_system_list() );

if ( !journal_debit_credit_difference_sum() )
{
        return (RECEIVABLE *)0;
}

RECEIVABLE_ACCOUNT_LIST *
        receivable_account_list_new(
                journal_system_list() );
```

RECEIVABLE_ACCOUNT_LIST

```
/* Usage */
RECEIVABLE_ACCOUNT_LIST *
    receivable_account_list_new(
        LIST *journal_system_list() );

/* Process */
if ( !list_rewind( journal_system_list ) )
{
    return (RECEIVABLE_ACCOUNT_LIST *)0;
}

RECEIVABLE_ACCOUNT_LIST *
    receivable_account_list_calloc(
        void );

LIST *list = list_new();

for JOURNAL *journal in journal_system_list
{
    RECEIVABLE_ACCOUNT *
        receivable_account =
            receivable_account_getset(
                list,
                journal->account_name );
    if ( journal->debit_amount )
    {
        receivable_account->debit_amount +=
            journal->debit_amount;
    }
    else
    {
        receivable_account->debit_amount -=
            journal->credit_amount;
    }
}
```



RECEIVABLE_ACCOUNT

```
/* Usage */
RECEIVABLE_ACCOUNT *
    receivable_account_getset(
        LIST *list,
        char *journal->account_name );

/* Process */
for RECEIVABLE_ACCOUNT *recievable_account in list
{
    if ( strcmp(
        account_name,
        receivable_account->account_name ) == 0 )
    {
        return receivable_account;
    }
}

RECEIVABLE_ACCOUNT *
    receivable_account =
        receivable_account_new(
            account_name );

RECEIVABLE_ACCOUNT *
    receivable_account_calloc(
        void );

list_set(
    list,
    receivable_account );

/* Attributes */
char *account_name;
double debit_amount;
```

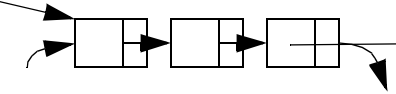
EMPLOYEE

char *full_name
char *street_address
double hourly_wage
double period_salary
double commission_sum_percent
double gross_pay_year_to_date
double net_pay_year_to_date
char * federal_marital_status
int federal_withholding_allowances
int federal_withholding_additional_period_amount
boolean state_withholding_exempt
char *state_marital_status
int state_withholding_allowances
int state_itemized_deduction_allowances
int retirement_contribution_plan_employee_period_amount
int retirement_contribution_plan_employer_period_amount
int health_insurance_employee_period_amount
int health_insurance_employer_period_amount
double union_dues_period_amount
char *terminated_date

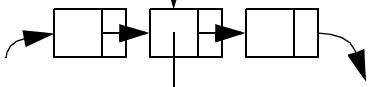
LIST *
employee_work_day_list

LIST *
employee_work_period_list

LIST *
customer_sale_list



CUSTOMER_SALE

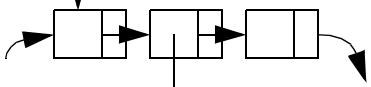


EMPLOYEE_WORK_PERIOD

char *full_name
char *street_address
int payroll_year
int payroll_period_number
char *begin_work_date
char *end_work_date
double regular_work_hours
double overtime_work_hours
double commission_sum
double gross_pay
double net_pay
double payroll_tax_amount
double federal_tax_withholding_amount
double state_tax_withholding_amount
double social_security_employee_tax_amount
double social_security_employer_tax_amount
double medicare_employee_tax_amout
double medicare_employer_tax_amount
int retirement_contribution_plan_employee_amount
int retirement_contribution_plan_employer_amount
int health_insurance_employee_amount
int health_insurance_employer_amount
double federal_unemployment_tax_amount
double state_unemployment_tax_amount
int union_dues_amount
char *transaction_date_time

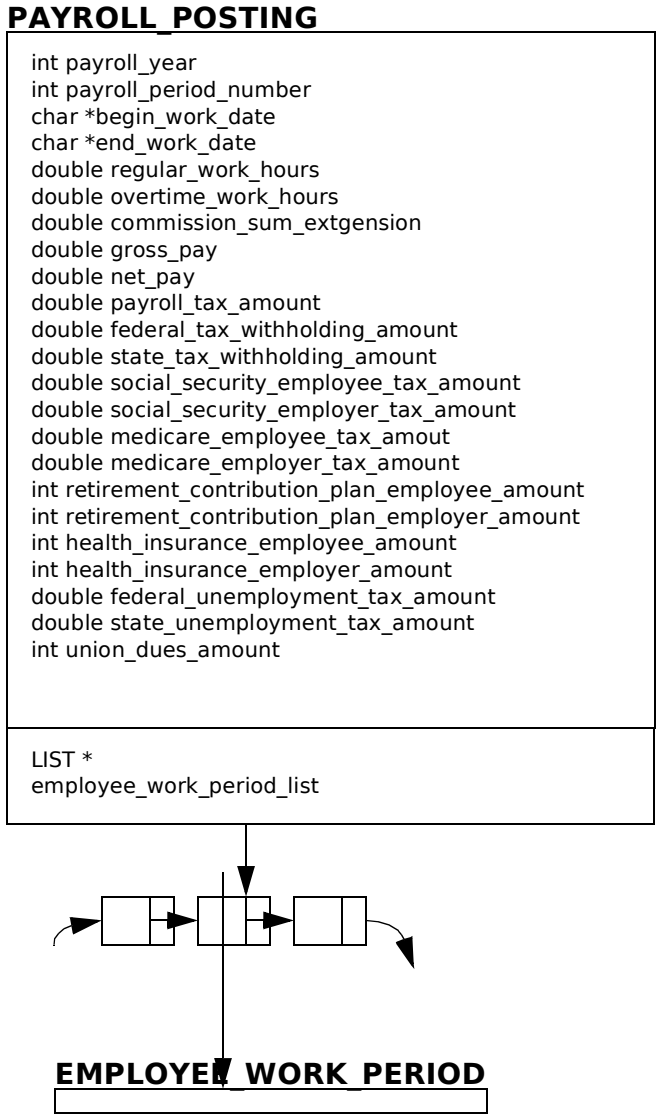
TRANSACTION *
transaction

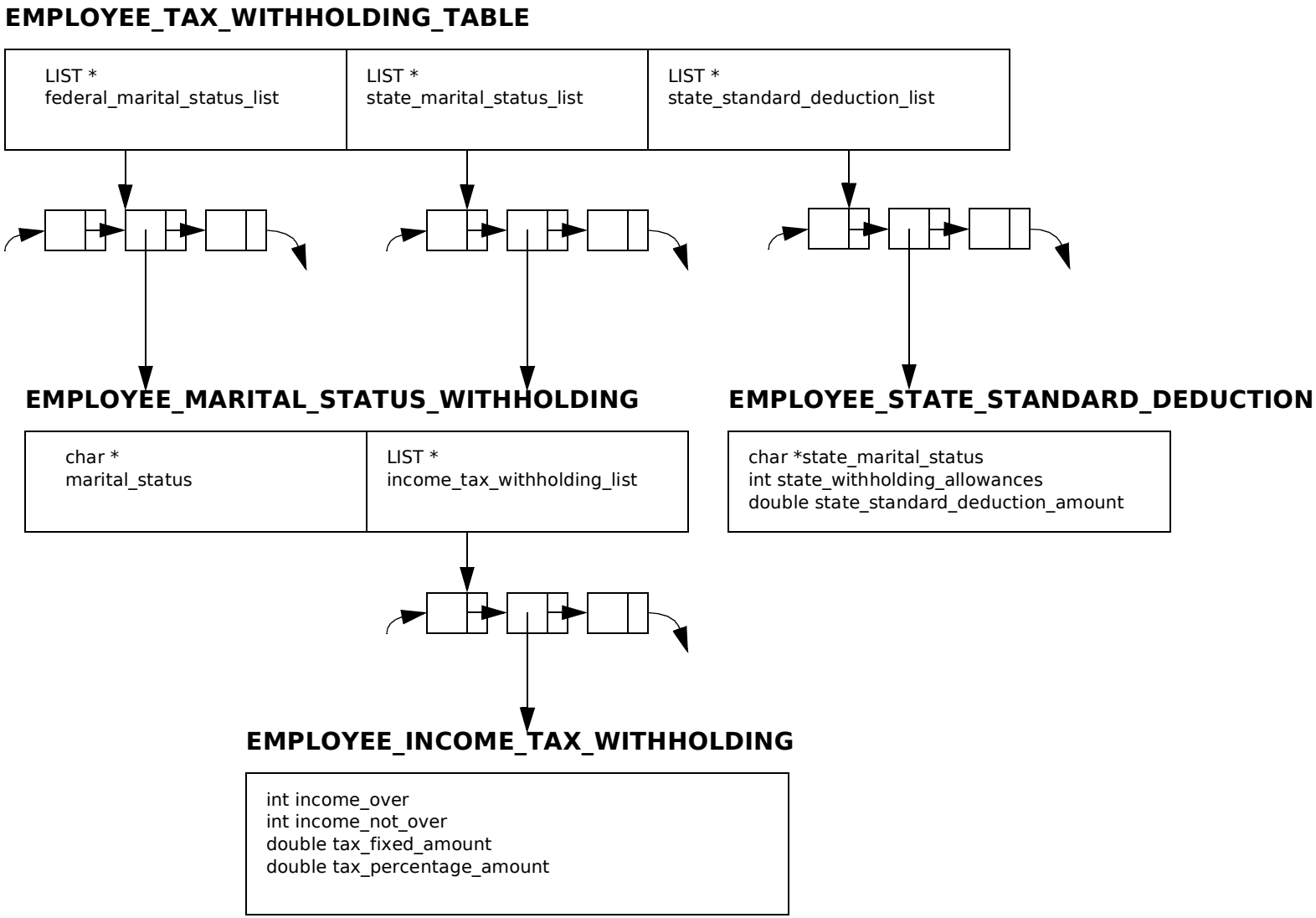
TRANSACTION



EMPLOYEE_WORK_DAY

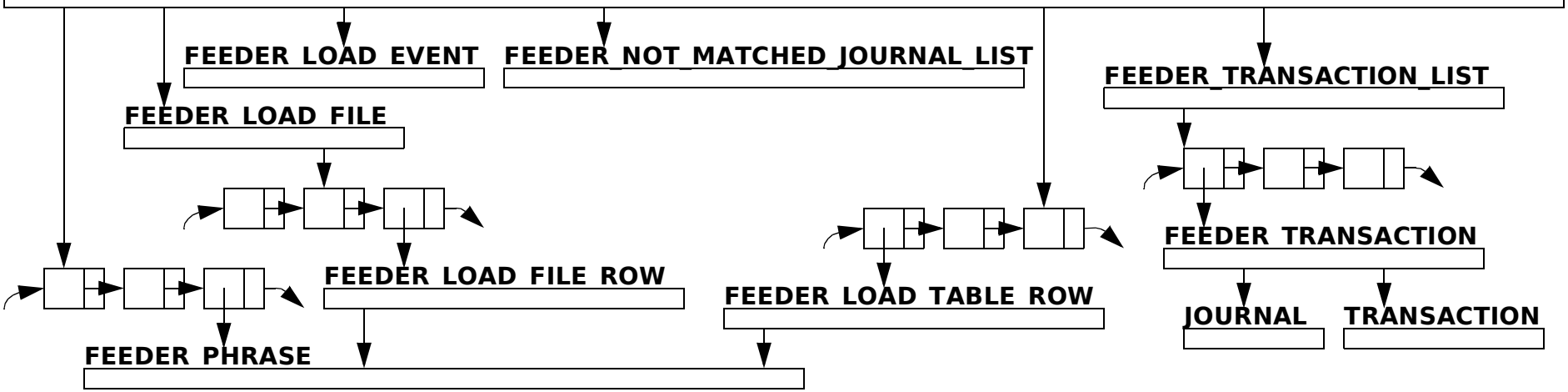
char *begin_work_date_time
char *end_work_date_time
double employee_work_hours
boolean overtime_work_day


PAYROLL_POSTING
int payroll_year
int payroll_period_number
char *begin_work_date
char *end_work_date
double regular_work_hours
double overtime_work_hours
double commission_sum_extgension
double gross_pay
double net_pay
double payroll_tax_amount
double federal_tax_withholding_amount
double state_tax_withholding_amount
double social_security_employee_tax_amount
double social_security_employer_tax_amount
double medicare_employee_tax_amout
double medicare_employer_tax_amount
int retirement_contribution_plan_employee_amount
int retirement_contribution_plan_employer_amount
int health_insurance_employee_amount
int health_insurance_employer_amount
double federal_unemployment_tax_amount
double state_unemployment_tax_amount
int union_dues_amount
LIST *
employee_work_period_list
EMPLOYEE_WORK_PERIOD


EMPLOYEE_TAX_WITHHOLDING_TABLE
LIST *
federal_marital_status_list
LIST *
state_marital_status_list
LIST *
state_standard_deduction_list
EMPLOYEE_MARITAL_STATUS_WITHHOLDING
char *
marital_status
LIST *
income_tax_withholding_list
EMPLOYEE_STATE_STANDARD_DEDUCTION
char *state_marital_status
int state_withholding_allowances
double state_standard_deduction_amount
EMPLOYEE_INCOME_TAX_WITHHOLDING
int income_over
int income_not_over
double tax_fixed_amount
double tax_percentage_amount

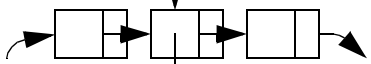
FEEDER_LOAD

```
/* Usage */
FEEDER_LOAD *feeder_load_new(
   char *process_name,
   char *login_name,
   char *feeder_account,
   char *feeder_load_filename,
   int date_column /* one_based */,
   int description_column /* one_based */,
   int debit_column /* one_based */,
   int credit_column /* one_based */,
   int balance_column /* one_based */,
   boolean reverse_order,
   double ending_balance );

/* Process */
FEEDER_LOAD *feeder_load_calloc( void );

LIST *feeder_phrase_list();

FEEDER_LOAD_FILE *feeder_load_file =
   feeder_load_file_fetch(
         char *feeder_load_filename,
         int date_column /* one_based */,
         int description_column /* one_based */,
         int debit_column /* one_based */,
         int credit_column /* one_based */,
         int balance_column /* one_based */,
         boolean reverse_order,
         LIST *feeder_phrase_list() );

char *feeder_load_file_minimum_date(
   reverse_order,
   feeder_load_file->feeder_load_file_row_list );

double feeder_load_row_last_running_balance(
   reverse_order,
   feeder_load_file->feeder_load_file_row_list );

char *feeder_load_basename_filename(
   char *feeder_load_filename );

char *feeder_load_sha256sum(
   char *feeder_load_filename );

FEEDER_LOAD_EVENT *feeder_load_event =
   feeder_load_event_new(
         date_display_19(
               DATE *date_now_new(
                     date_utc_offset()
                           /* feeder_load_date_time */ ) ),
         login_name,
         feeder_account,
         feeder_load_basename_filename(),
         ending_balance,
         feeder_load_row_last_running_balance(),
         feeder_load_sha256sum() );

FEEDER_NOT_MATCHED_JOURNAL_LIST *
   feeder_not_matched_journal_list =
         feeder_not_matched_journal_list_fetch(
               feeder_account,
               feeder_load_file->
                     feeder_load_file_row_list );

LIST *feeder_load_table_row_list(
         feeder_account,
         feeder_phrase_list(),
         feeder_load_file_minimum_date() );

FEEDER_TRANSACTION_LIST *
   feeder_transaction_list =
         feeder_transaction_list_new(
               feeder_account,
               feeder_load_file->feeder_load_row_list,
               feeder_phrase_list,
               feeder_journal_list->journal_orphan_list,
               feeder_load_row_existing_list );

double feeder_load_prior_running_balance(
   char *feeder_account,
   char *feeder_load_file_minimum_date() );
```



FEEDER_NOT_MATCHED_JOURNAL_LIST

```
/* Usage */
FEEDER_NOT_MATCHED_JOURNAL_LIST *
    feeder_journal_list_new(
        char *feeder_account,
        LIST *feeder_load_file_row_list );

/* Process */
FEEDER_NOT_MATCHED_JOURNAL_LIST *
    feeder_not_matched_journal_list_calloc(
        void );

char *feeder_not_matched_journal_list_subquery(
    char *JOURNAL_TABLE,
    char *FEEDER_LOAD_ROW_TABLE );

char *feeder_not_matched_journal_list_where(
    char *feeder_account,
    char *feeder_not_matched_journal_list_subquery() );

LIST *journal_list =
    journal_system_list(
        char *journal_system_string(
            JOURNAL_SELECT,
            JOURNAL_TABLE,
            feeder_not_matched_journal_list_where() ),
        1 /* fetch_check_number */,
        0 /* not fetch_memo */ );

LIST *list = list_new();

for JOURNAL *journal in journal_list
{
    list_set(
        list,
        FEEDER_JOURNAL *feeder_journal_new(
            feeder_account,
            feeder_load_file_row_list,
            journal ) );
}
```



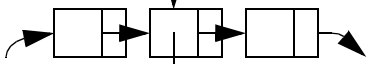
FEEDER_JOURNAL

```
/* Usage */
FEEDER_JOURNAL *feeder_journal_new(
    char *feeder_account,
    LIST *feeder_load_file_row_list,
    JOURNAL *journal );

/* Process */
FEEDER_JOURNAL *feeder_journal_calloc(
    void );

if ( journal->check_number )
{
    FEEDER_LOAD_FILE_ROW *
        feeder_load_file_row =
            feeder_load_file_check_number_seek(
                journal->check_number,
                feeder_load_file_row_list );
}

if ( !feeder_load_file_row )
{
    FEEDER_LOAD_FILE_ROW *
        feeder_load_file_row =
            feeder_load_file_row_amount_seek(
                journal->debit_amount,
                journal->credit_amount,
                feeder_load_file_row_list );
}

if ( feeder_load_file_row )
{
    feeder_load_file_row_taken();
}
```

JOURNAL **FEEDER_LOAD_FILE_ROW**

FEEDER_TRANSACTION_LIST

```
/* Usage */
FEEDER_TRANSACTION_LIST *
   feeder_transaction_list_new(
         char *feeder_account,
         LIST *feeder_load_file->feeder_load_row_list,
         LIST *feeder_phrase_list,
         LIST *feeder_journal_list->journal_orphan_list,
         LIST *feeder_load_row_existing_list );

/* Process */
FEEDER_TRANSATION_LIST *
   feeder_transaction_list_calloc(
         void );

LIST *list = list_new();

for FEEDER_LOAD_ROW *feeder_load_row in
   feeder_load_row_list
{
   list_set(
      list,
      FEEDER_TRANSACTION *feeder_transaction_new(
            feeder_account,
            feeder_load_row,
            feeder_phrase_list,
            journal_orphan_list,
            feeder_load_row_existing_list ) );
   }
}
```



FEEDER_TRANSACTION

```
/* Usage */
FEEDER_TRANSACTION *
   feeder_transaction_new(
         char *feeder_account,
         FEEDER_LOAD_FILE_ROW *feeder_load_file_row,
         LIST *feeder_phrase_list,
         LIST *journal_orphan_list,
         LIST *feeder_load_row_existing_list );

/* Process */
FEEDER_TRANSATION *feeder_transaction_calloc(
   void );
```

FEEDER_LOAD_FILE_ROW **JOURNAL** **TRANSACTION**

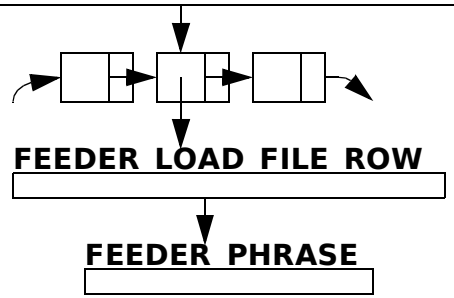
FEEDER_PHRASE

FEEDER_LOAD_FILE

```
/* Usage */
FEEDER_LOAD_FILE *
    feeder_load_file_fetch(
        char *feeder_load_filename,
        int date_column /* one_based */,
        int description_column /* one_based */,
        int debit_column /* one_based */,
        int credit_column /* one_based */,
        int balance_column /* one_based */,
        boolean reverse_order,
        LIST *feeder_phrase_list() );

/* Process */
FEEDER_LOAD_FILE *feeder_load_file_calloc(
    void );

int feeder_load_file_line_count(
    char *feeder_load_filename,
    int date_column - 1 /* date_piece_offset */ );

int feeder_load_file_begin_sequence_number(
    int feeder_load_file_line_count() );
```

```
FILE *file = fopen( feeder_load_filename, "r" );

LIST *feeder_load_file_row_list = list_new();
int line_number = 0;

for char *input in string_input( file )
{
    timlib_remove_character( input, '\\' );

    FEEDER_LOAD_FILE_ROW *feeder_load_file_row =
        feeder_load_file_row_new(
            date_column /* one_based */,
            description_column /* one_based */,
            debit_column /* one_based */,
            credit_column /* one_based */,
            balance_column /* one_based */,
            feeder_phrase_list(),
            input,
            ++line_number );

    if ( !feeder_load_file_row ) continue;
```

```
    if ( reverse_order )
    {
        list_set_first(
            feeder_load_file_row_list,
            feeder_load_file_row );
    }
    else
    {
        list_set(
            feeder_load_file_row_list,
            feeder_load_file_row );
    }
}

fclose( file );

/* Public */
char *feeder_load_file_minimum_date(
    boolean reverse_order,
    LIST *feeder_load_file_row_list );
```



FEEDER_LOAD_FILE_ROW (1)

```
/* Usage */
FEEDER_LOAD_FILE_ROW *feeder_load_file_row_new(
	int date_column /* one_based */,
	int description_column /* one_based */,
	int debit_column /* one_based */,
	int credit_column /* one_based */,
	int balance_column /* one_based */,
	LIST *feeder_phrase_list(),
	char *input,
	int line_number );

/* Process */
int feeder_load_file_row_line_number(
	int line_number );

char *american_date =
	strdup(
		piece_quote_comma(
			buffer,
			input,
			date_column - 1 ) );

char *feeder_load_file_row_international_date(
	char *american_date );

if ( !feeder_load_file_row_international_date() )
{
	return (FEEDER_LOAD_FILE_ROW *)0;
}

char *description =
	strdup(
		piece_quote_comma(
			buffer,
			input,
			description_column - 1 ) );

if ( !credit_column )
{
	double amount =
		atof(
			piece_quote_comma(
				buffer,
				input,
				debit_column - 1 ) );
}
else
{
	double debit =
		atof(
			piece_quote_comma(
				buffer,
				input,
				debit_column - 1 ) );

	double credit =
		atof(
			piece_quote_comma(
				buffer,
				input,
				credit_column - 1 ) );

	double amount =
		feeder_load_row_amount(
			debit,
			credit );
}

if ( balance_column )
{
	double balance =
		atof(
			piece_quote_comma(
				buffer,
				input,
				balance_column - 1 ) );
}

int feeder_load_file_row_check_number(
	char *description );

char *feeder_load_file_row_description_embedded(
	description,
	balance,
	line_number,
	feeder_load_file_row_check_number() );

/* Usage */
char *feeder_load_file_row_description_embedded(
	char *description,
	double balance,
	int line_number,
	int feeder_load_file_row_check_number() );

/* Process */
if ( feeder_load_file_row_check_number() )
{
	return description;l
}

char *feeder_load_file_row_trim_date(
	char *description );

char *sed_trim_double_spaces(
	feeder_load_file_row_trim_date() );

char *feeder_load_file_row_description_build(
	char *sed_trim_double_spaces(),
	double balance,
	int line_number );

char *feeder_load_file_row_description_crop(
	char *feeder_load_file_row_description_build(),
	int FEEDER_DESCRIPTION_SIZE );

FEEDER_PHRASE *
	feeder_phrase_seek(
		char *feeder_load_file_row_international_date(),
		char *feeder_load_file_row_description_crop(),
		LIST *feeder_phrase_list() );
```

FEEDER_PHRASE

FEEDER_LOAD_FILE_ROW (2)

```
/* Usage */
void feeder_load_file_row_list_insert(
      char *FEEDER_LOAD_ROW_SELECT,
      char *FEEDER_LOAD_TABLE,
      LIST *feeder_load_row_list );

/* Process */
char *feeder_load_file_row_list_insert_system_string(
      char *FEEDER_LOAD_ROW_SELECT,
      char *FEEDER_LOAD_ROW_TABLE );

FILE *feeder_load_file_row_list_insert_pipe(
      feeder_load_file_row_list_insert_system_string() );

/* Private */
void feeder_load_file_row_insert(
      FILE *feeder_load_file_row_list_insert_pipe(),
      char *feeder_load_event->feeder_load_date_time,
      char *full_name,
      char *street_address,
      char *transaction_date_time,
      char *feeder_account,
      char *feeder_load_file_row_international_date(),
      char *feeder_load_file_row_description_embedded(),
      double feeder_load_file_row->amount,
      double feeder_load_file_row->balance,
      int row_number );

/* Public */
double feeder_load_file_row_last_running_balance(
      boolean reverse_order,
      LIST *feeder_load_row_list );

FEEDER_LOAD_FILE_ROW *
      feeder_load_file_row_check_number_seek(
            int check_number,
            LIST *feeder_load_file_row_list );

FEEDER_LOAD_FILE_ROW *
      feeder_load_file_row_amount_seek(
            double debit_amount,
            double credit_amount,
            LIST *feeder_load_file_row_list );

boolean feeder_load_file_row_taken( void );
```

FEEDER_LOAD_TABLE_ROW

```
/* Usage */
LIST *feeder_load_table_row_list(
    char *feeder_account,
    LIST *feeder_phrase_list(),
    char *feeder_load_file_minimum_date() );

/* Process */
char *feeder_load_table_row_where(
    char *feeder_account,
    char *feeder_load_file_minimum_date );

char *feeder_load_table_row_system_string(
    char *FEEDER_LOAD_TABLE_ROW_SELECT,
    char *FEEDER_LOAD_ROW_TABLE,
    char *feeder_load_table_row_where() );

LIST *list = list_new()

FILE *input_pipe =
    popen(  feeder_load_table_row_system_string(),
            "r" );

while ( input = string_input( input_pipe ) )
{
    list_set(
        list,
        FEEDER_LOAD_TABLE_ROW *
            feeder_load_table_row_parse(
                feeder_phrase_list(),
                input ) );
}

pclose( input_pipe );

/* Usage */
FEEDER_LOAD_TABLE_ROW *
    feeder_load_table_row_parse(
        LIST *feeder_phrase_list(),
        char *input );

/* Process */
FEEDER_LOAD_TABLE_ROW *
    feeder_load_table_row_calloc(
        void );

FEEDER_PHRASE *feeder_phrase =
    feeder_phrase_seek(
        char *feeder_date,
        char *feeder_description,
        LIST *feeder_phrase_list() );

/* Attributes */
char *feeder_date;
char *feeder_description;
char *transaction_date_time;
FEEDER_PHRASE *feeder_phrase;
```

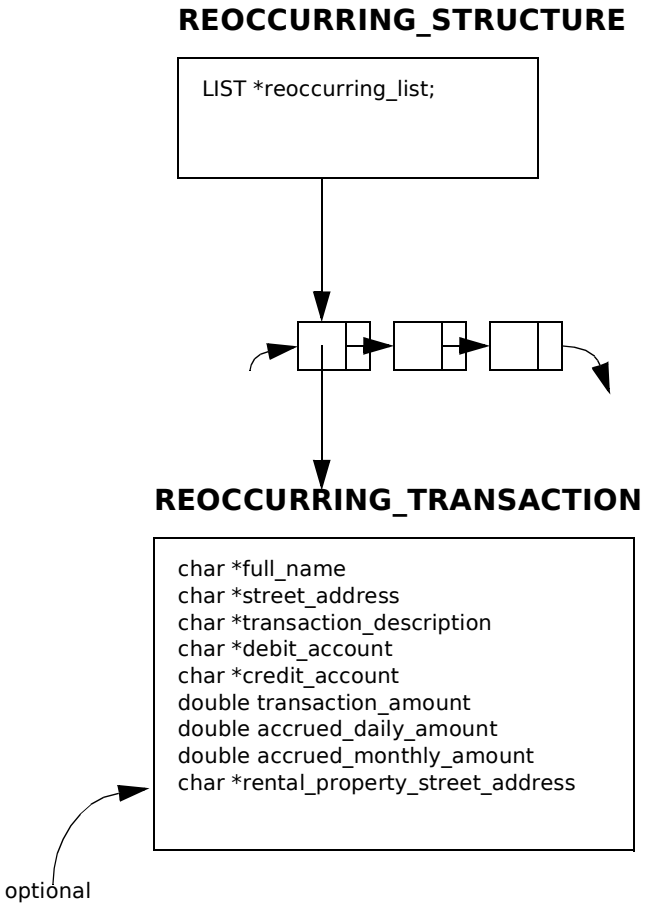
FEEDER_PHRASE

FEEDER_PHRASE

```
/* Usage */
LIST *feeder_phrase_list( void );

/* Process */
char *feeder_phrase_system_string(
	char *FEEDER_PHRASE_SELECT,
	char *FEEDER_PHRASE_TABLE );

FILE *input_pipe =
	popen(	feeder_phrase_system_string(),
		"r" );

LIST *list = list_new();

while ( input = string_input( input_pipe ) )
{
	list_set(
		list,
		feeder_phrase_parse( input ) );
}

/* Usage */
FEEDER_PHRASE *feeder_phrase_parse(
	char *input );

/* Process */
FEEDER_PHRASE *feeder_phrase_new(
	char *feeder_phrase );

FEEDER_PHRASE *feeder_phrase_calloc( void );

/* Attributes */
char *feeder_phrase;
char *nominal_account;
char *full_name;
char *street_address;
boolean feeder_phrase_ignore;

/* Public */
FEEDER_PHRASE *feeder_phrase =
	feeder_phrase_seek(
		char *feeder_date,
		char *feeder_description,
		LIST *feeder_phrase_list() );
```

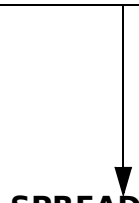
FEEDER_LOAD_EVENT

```
/* Usage */
FEEDER_LOAD_EVENT *feeder_load_event_new(
	char *feeder_load_date_time,
	char *login_name,
	char *feeder_load_basename_filename(),
	char *feeder_account,
	double ending_balance,
	double feeder_load_row_last_running_balance(),
	char *feeder_load_sha256sum() );

/* Process */
FEEDER_LOAD_EVENT *feeder_load_event_calloc(
	void );

double feeder_load_event_ending_balance(
	double ending_balance,
	double feeder_load_row_last_running_balance );

/* Attributes */
char *feeder_load_date_time;
char *login_name;
char *basename_filename;
char *feeder_account;
double ending_balance;
char *sha256sum;

/* Driver */
void feeder_load_event_insert(
	char *FEEDER_LOAD_EVENT_TABLE,
	char *FEEDER_LOAD_SELECT,
	char *feeder_load_date_time,
	char *login_name,
	char *basename_filename,
	char *sha256sum,
	char *feeder_account,
	double ending_balance );

/* Process */
char *feeder_load_event_insert_system_string(
	char *FEEDER_LOAD_EVENT_TABLE,
	char *FEEDER_LOAD_SELECT,
	char SQL_DELIMITER );

FILE *insert_pipe =
	popen(	feeder_load_event_insert_system_string(),
		"w" );

void feeder_load_event_insert_pipe(
	FILE *insert_pipe,
	char SQL_DELIMITER,
	char *feeder_load_date_time,
	char *login_name,
	char *basename_filename,
	char *sha256sum,
	char *feeder_account,
	double ending_balance );

pclose( insert_pipe );
```


REOCCURRING_STRUCTURE
LIST *reoccurring_list;
REOCCURRING_TRANSACTION
char *full_name
char *street_address
char *transaction_description
char *debit_account
char *credit_account
double transaction_amount
double accrued_daily_amount
double accrued_monthly_amount
char *rental_property_street_address
optional

PAYPAL

```
/* Usage */
PAYPAL *paypal_fetch(
      spreadsheet_filename,
      date_label );

/* Process */
SPREADSHEET *spreadsheet =
      spreadsheet_fetch(
            spreadsheet_filename,
            date_label );
```



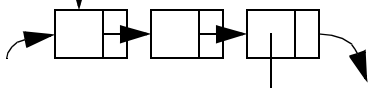
SPREADSHEET

PAYPAL_DATASET

```
/* Input */
char *date_A;
char *time_B;
char *timezone_C;
char *full_name_D;
char *transaction_type_E;
char *transaction_status_F;
char *currency_G;
char *gross_revenue_H;
char *transaction_fee_I;
char *net_revenue_J;
char *from_email_address_K;
char *to_email_address_L;
char *transaction_ID_M;
char *shipping_address_N;
char *address_status_O;
char *item_title_P;
char *item_ID_Q;
char *shipping_handling_amount_R;
char *insurance_amount_S;
char *sales_tax_T;
char *option_1_name_U;
char *option_1_value_V;
char *option_2_name_W;
char *option_2_value_X;
char *reference_txn_ID_Y;
char *invoice_number_Z;
char *custom_number_AA;
char *quantity_AB;
char *receipt_ID_AC;
char *balance_AD;
char *address_line_1_AE;
char *address_2_district_neighborhood_AF;
char *town_city_AG;
char *state_province_AH;
char *zip_code_AI;
char *country_AJ;
char *contact_phone_number_AK;
char *subject_AL;
char *note_AM;
char *country_code_AN;
char *balance_impact_AO;
```

SPREADSHEET

```
/* Input */
char *spreadsheet_filename;
char *date_label;

/* Process */
LIST *spreadsheet_column_list(
      spreadsheet_filename,
      date_label );

char *maximum_date;

char *spreadsheet_minimum_date(
      &maximum_date,
      spreadsheet_filename );

char *spreadsheet_heading_data(
      spreadsheet_column_list(),
      input_row,
      heading );
```



SPREADSHEET_COLUMN

```
/* Usage */
SPREADSHEET_COLUMN *spreadsheet_column_new(
      heading,
      piece_offset );

/* Process */
SPREADSHEET_COLUMN *spreadsheet_column_heading_seek(
      spreadsheet_column_list,
      heading );
```

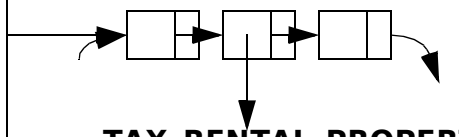
SPREADSHEET_UPLOAD_EVENT

```
/* Input */
char *spreadsheet_filename;

/* Process */
char *spreadsheet_upload_date_time =
      date_time19_now();

char *spreadsheet_upload_file_sha256sum(
      spreadsheet_filename );
```

TAX

```
/* Usage */
TAX *tax_new(
      char *tax_form_string,
      int tax_year );

/* Input */
char *tax_form_string;
int tax_year;

/* Attributes */
```

```
TAX_FORM *tax_form =
      tax_form_fetch();

RECOVERY *recovery;

LIST *tax_rental_property_list =
      tax_rental_property_list_fetch(
            tax_form_string,
            tax_year );
```



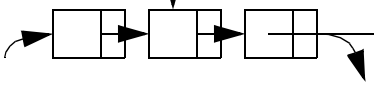
TAX_RENTAL_PROPERTY

TAX_FORM

```
/* Usage */
TAX_FORM *tax_form_fetch(
      char *tax_form_string,
      int tax_year,
      boolean fetch_line_list,
      boolean fetch_account_list,
      boolean fetch_journal_list );

/* Attributes */
char *tax_form_string;

/* Input */
int tax_year;

/* Process */
LIST *tax_form_line_list(
      tax_form_string,
      tax_year,
      fetch_account_list,
      fetch_journal_list );
```

RECOVERY

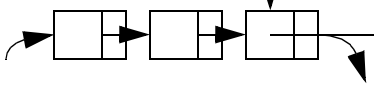
```
double fixed_asset_recovery_amount;
double prior_fixed_asset_recovery_amount;
double property_recovery_amount;
double prior_property_recovery_amount;
double total_recovery_amount;
```



TAX_FORM_LINE

```
/* Usage */
LIST *tax_form_line_list(
      char *tax_form_string,
      int tax_year,
      boolean fetch_account_list,
      boolean fetch_journal_list );

/* Attributes */
char *tax_form_string;
char *tax_form_line_string;
char *tax_form_description;
boolean itemize_accounts;

/* Input */
int tax_year;
```

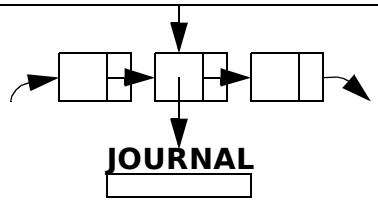
```
/* Process */
LIST *tax_form_line_account_list(
      tax_form_string,
      tax_form_line_string,
      tax_year,
      fetch_journal_list );

/* Operations */
void tax_form_line_list_steady_state(
      tax_form_line_list );

double tax_form_line_total(
      tax_form_line_account_list() );
```



TAX_FORM_LINE_ACCOUNT

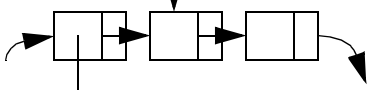
TAX_FORM_LINE_ACCOUNT

```
/* Usage */
LIST *tax_form_line_account_list(
    char *tax_form_string,
    char *tax_form_line,
    int tax_year,
    boolean fetch_journal_list );

/* Attributes */
char *tax_form_string;
char *tax_form_line;
char *account_name;

/* Process */
ACCOUNT *account =
    account_fetch(
        account_name,
        1 /* fetch_subclassification */,
        1 /* fetch_entity */ );

if ( fetch_journal_list )
{
    boolean subclassification_current_liability(
        account->subclassification_name );

    LIST *journal_year_list(
        tax_year,
        account_name,
        1 /* fetch_memo */ );

    double tax_form_line_account_total(
        journal_year_list(),
        account->
            subclassification->
            entity->
            accumulate_debit,
        subclassification_current_liability() );
}
```



TAX_RENTAL_PROPERTY

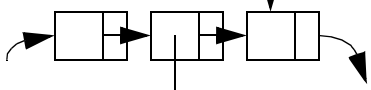
```
/* Usage */
LIST *tax_rental_property_list_fetch(
      char *tax_form_string,
      int tax_year );

/* Attributes */
char  *property_street_address;

/* Process */
LIST *tax_rental_property_string_list(
      tax_rental_property_list );

LIST *tax_form_line_rental_list(
      tax_form_line_list(),
      tax_rental_property_string_list );

double recovery_amount;
double revenue_total [23a];
double mortgage_interest_total [23c];
double depreciation_total [23d];
double expense_total [23e];
double positive_net_income [24];
double loss_total [25];
double tax_rental_property_total();
```



TAX_FORM_LINE_RENTAL

```
char *tax_form_line;
char *tax_form_description;

enum tax_form_line_category;
LIST *tax_form_line_account_list();

LIST *rental_property_list;
double tax_form_line_rental_property_total();
```

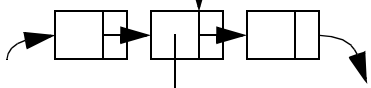
revenue
mortgage_interest_paid
depreciation_expense
operating_expense
loss



TAX_FORM_LINE_ACCOUNT

INVESTMENT_ACCOUNT

```
ENTITY *financial_entity =
      entity_new(
            full_name,
            street_address );

char *account_number;
char *investment_classification;
char *investment_purpose;

LIST *investment_account_balance_list(
full_name,
street_address,
account_number );
```

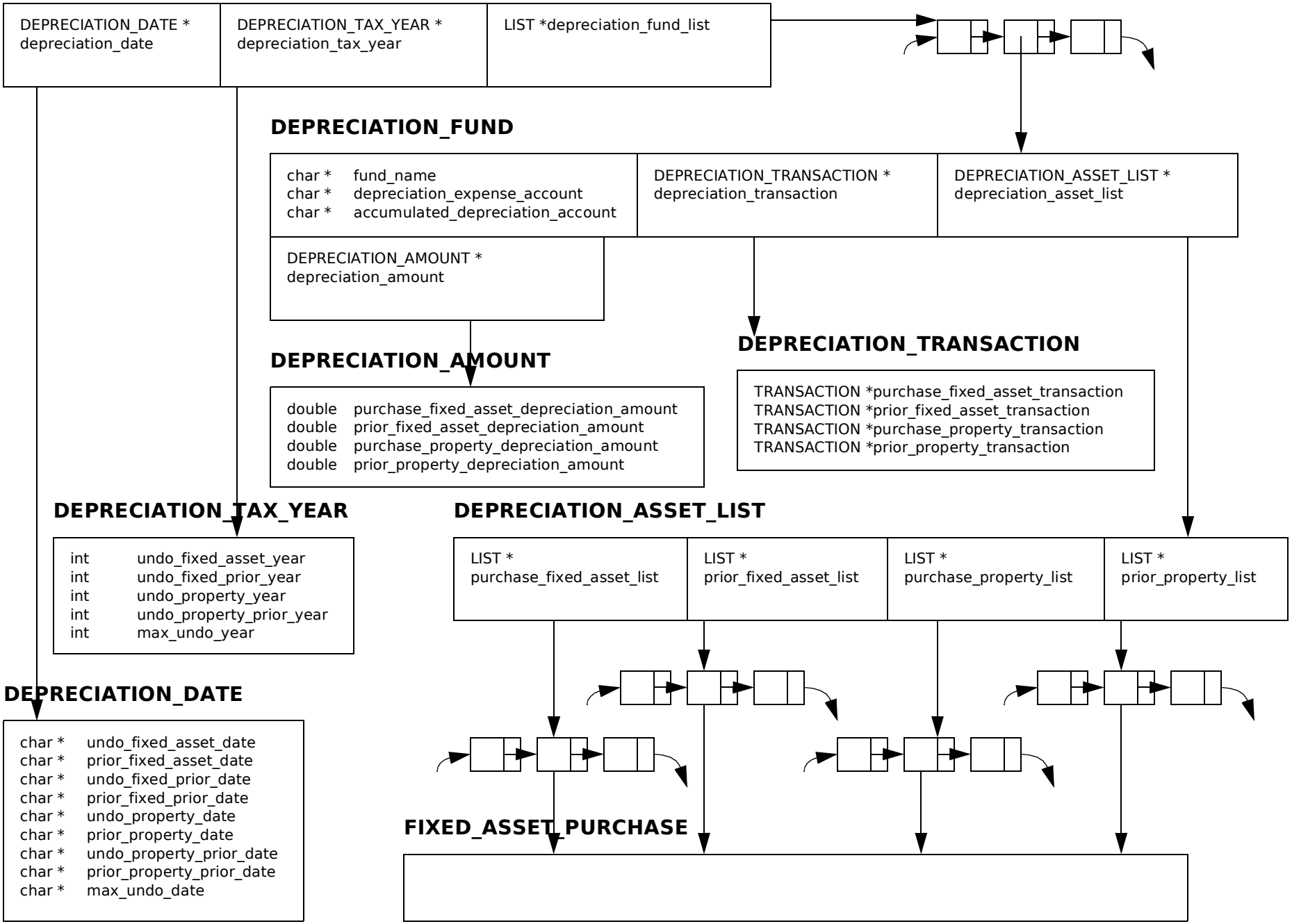


ACCOUNT_BALANCE

```
char *date
double balance;
double balance_change;
int balance_change_percent;
```

TRANSACTION

ACCOUNT_BALANCE_EQUITY

```
char *date_time
char *investment_operation
double share_price
doubel share_quantity_change
double share_quantity_balance
double cash_in
double market_value
double book_value_change
double book_value_balance
double moving_share_price
double unrealized_gain_balance
double unrealized_gain_change
double reaqlized_gain
boolean table_first_row
char *transaction_date_time
TRANSACTION *
transaction
```

DEPRECIATION_STRUCTURE